

Thesis for the Degree of Master of Arts

Conjoined *wh*-sluicing in English:
A corpus-based approach

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Abstract

This thesis investigates the phenomenon of conjoined *wh*-sluicing in English, a type of elliptical construction that exhibits unique properties compared to other multiple *wh*-constructions, such as multiple sluicing and coordinated multiple *wh*-questions. Conjoined *wh*-sluicing demonstrates insensitivity to locality restrictions, distinguishing it as a unique elliptical linguistic phenomenon. Semantically, conjoined *wh*-sluicing limits the scope of its correlates, differentiating it from multiple sluicing.

Previous analyses within the Minimalist framework—including mono-clause, reverse sluicing, bi-clausal bulk sharing, and bi-clausal non-bulk sharing—fail to fully account for certain instances of conjoined *wh*-sluicing. To address this, a corpus investigation utilizing six corpora from the BNC corpus family was conducted, revealing several key findings.

The discussion section addressed idiosyncratic instances of the corpus dataset, suggesting the need for new approaches to the construction. The authentic data highlighted some basic properties of the construction followed by a brief qualitative analysis.

Ultimately, this thesis proposes a Construction Grammar (CxG) and Head-Driven Phrase Structure Grammar (HPSG) approach to conjoined *wh*-sluicing in English, which addresses syntactic identity issues observed in the corpus data, while also accounting for the semantic and pragmatic aspects of the construction by considering the dialogue, positing that this approach will be able to explain the syntactic identity issues of this construction.

Keywords: sluicing, multiple sluicing, coordination, ellipsis, Construction-based Grammar

List of Abbreviations

The following is a list of abbreviations used in glossing examples.

AdvP	Adverb Phrase
ARG-ST	Argument-Structure
AVM	Attributed-Value Matrices
CNPC	Complex Noun Phrase Constraint
CxG	Construction Grammar
DGB	Dialogue Game Board
DI	Direct Interpretation
FEC	Focus-Establishing Constituent
HPSG	Head-driven Phrase Structure Grammar
MAX-QUD	Maximal Question Under Discussion
NP	Noun Phrase
NSU	Non-Sentential Utterance
QUD	Question Under Discussion
S	Sentence
SAL-UTT	Salient Utterance
VP	Verb Phrase

Chapter 1

Introduction

Sluicing is an elliptical construction where everything except a single *wh*-expression is elided from the structure and the interpretation of the construction relies on the preceding context. This linguistic phenomenon has been analyzed crosslinguistically in abundant amount of previous papers (Ross 1969; Chung et al. 1995; Giannakidou and Merchant 1998; Ginzburg and Sag 2000; Merchant 2001, 2003; Chung 2013; Kim 2013, 2015, among others). Sluicing consists of an antecedent which can either have an overt or covert correlate, which aids the interpretation of the elided site after the *wh*-remnant. Sluicing has two types: sprouting, which does not have an overt correlate in the antecedent, and merger type, which has an overt correlate in the antecedent.

- (1) a. He's talking, but you can't imagine **where/why/how/how fast/with whom**.

(Adapted from Chung et al. 1995: (3a)) [Sprouting]

- b. He's looking at something. I can't imagine **what**.

(Adapted from Chung et al. 1995: (4b)) [Merger]

While the remnants in (1a) do not have any overt correlates, (1b) does, which is the underlined '*something*' in the antecedent clause. One thing to note is that overt correlates consist of indefinite nouns, for example, noun phrases with the determiner *a/an*, quantified pronouns beginning with *some*, an indefinite pronoun '*one*', and such.

Multiple *wh*-constructions are productive within languages that allow multiple *wh*-fronting,

such as Hungarian, Bulgarian, Serbo-Croatian, Russian, and Romanian (Rudin, 1988; Boskovic, 2002; Grebenyova, 2004; Lasnik, 2007; Grebenyova, 2009; Lasnik, 2014; Ronai and Stigliano, 2021). Among these constructions, multiple sluicing, where more than one *wh*-phrases are able to undergo sluicing, is acceptable within those languages. Since English is a non-multiple-*wh* fronting language, the productivity of multiple *wh*-words in a construction with multiple *wh*-movement is restricted and there can be issues raised regarding its syntactic well-formedness as well.

- (2) a. Valaki meghívott valakit, de nem tudom ki kit
 someone invited someone.ACC but not know.I who.NOM who.ACC
 ‘Someone invited someone, but I don’t know who whom.’
 (Ronai and Stigliano 2021: (1)) [Hungarian]
- b. Njakoј e vidjal njakogo, no ne znam koj kogo.
 someone AUX seen someone but not I.know who whom
 ‘Someone saw someone, but I don’t know who saw whom.’
 (Merchant 2003: (56b)) [Bulgarian]
- (3) a. ?* Someone saw something, but I don’t know **who what**. (Lasnik 2007: (34))
- b. ? Someone talked about something, but I don’t know **who about what**. (Lasnik 2007: (35))
- c. Some boy likes some girl, but I can’t remember **which boy which girl**.
 (Kotek and Barros 2018: (2c))

While (2) holds examples of some languages where multiple sluicing is felicitous, but observing (3), the use of multiple sluicing in the English language regarding its *wh*-remnant types that could be sluiced. The remnant pair that consists of a *wh*-remnant and another *wh*-remnant following a preposition is considered to be marginally acceptable but a pair with ‘*which* + noun’ as both of the remnants is regarded as acceptable. Notice, that the acceptability of multiple sluicing improves regardless of the types of remnant pairs when a coordinating conjunction *and* is inserted between the two remnants.

- (4) a. Someone saw something, but I don't know **who** and **what**. (Adapted from [Lasnik 2007](#): (34))
- b. I remember that someone asked about something in class, but I don't recall **who** and **about what**. (Adapted from [Lasnik 2007](#): (35))
- c. A boy kissed a girl, but I can't remember exactly **which boy** and **which girl**. (Adapted from [Kotek and Barros 2018](#): (2c))

I would like to refer to this linguistic phenomenon as conjoined *wh*-sluicing, where a coordinating conjunction conjoins the two *wh*-remnants in the multiple sluicing construction and in doing so, repairs the grammaticality of the construction. Given that the simple presence of ‘*and*’ allows multiple sluicing to become felicitous, this thesis aims to account for some basic properties of conjoined *wh*-sluicing, observe the analytical approaches to the construction, then try to display the actual usage of the construction with data collected from various corpora, and discuss the findings from the corpus search as well as raising some questions to the analyses given previously from the data analysis, finally proposing a novel viewpoint to the conjoined *wh*-sluicing phenomenon that can also explain the idiosyncratic cases drawn from the collected corpus data.

Chapter 2

Properties of conjoined *wh*-sluicing

2.1 Syntactic properties

Conjoined *wh*-sluicing in English shows distinct properties that are unique in comparison with other multiple *wh*-constructions such as multiple sluicing and coordinated multiple *wh*-questions. The comparison between conjoined *wh*-sluicing with these other constructions will demonstrate that conjoined *wh*-sluicing is not derived from another construction, and is a discrete construction of its own.

2.1.1 Argument structure requirement

Sluicing must follow the argument structure condition where the argument structure of the putative source clause of the sluiced site should correspond to that of the antecedent, and also inherit its contextual information as well (Chung et al., 1995; Romero, 1998; Merchant, 2001; Aelbrecht, 2010; Chung, 2013). While (1a) satisfies the argument structure required by the passive construction with the remnant ‘*by whom*,’ the condition is not fulfilled in (1b), and therefore its acceptability can be questioned.

- (1) a. The jewel was stolen, but I can’t remember **by whom**.
- b. ?? The jewel was stolen, but I can’t remember **who**. (Chung 2013: 34)

- (2) a. The jewel was stolen, but we can't remember **by whom and why**.
 b. ?? The jewel was stolen, but we can't remember **who and why**. (Adapted from [Chung 2013](#): 34)

This condition is also applied to conjoined *wh*-sluicing, where the argument structure generated by the passivized structure in the antecedent should parallel with the argument structure of the putative source clause of the remnant. The interpreted meaning of the elided site based on this condition is applied both of the *wh*-remnants ([Chung et al., 1995](#)).

This argument structure requirement is also related to the two types of sluicing, merger and sprouting. While merger refers to some syntactic element (correlate) in the antecedent clause, sprouting refers to a *wh*-remnant that is not quite mentioned in the preceding clause, but can be implied from the given context ([Chung et al. 1995](#), [Merchant 2001](#), and so on).

- (3) a. Lee called, but I don't know **when** <he called>. [Sprouting]
 b. Lee called someone, but I don't know **whom** <he called>.
 (Adapted from [Merchant 2001](#): (2a)) [Merger]

The 'someone' in the antecedent clause in (3b) serves as a correlate of the remnant *whom*, and is a merger case. However, in (3a), the sentence shows no correlate in the antecedent for the adjunct *when*, so it is a sprouting case. This remnant *when* is used to elaborate on the time or the period at which Lee called. Even though these two are distinct types of sluicing, the argument structure required for both are identical. This process is also applied to the conjoined *wh*-sluicing, even though it has multiple remnants.

- (4) a. Lee called someone, but I don't know **whom and when**.
 b. Lee called someone, but I don't know **whom** <Lee called> **and when** <Lee called>. (Adapted from [Merchant 2001](#): (2a))

Here, the two types can be conjoined together and share the identical argument structure requirements, for its antecedent clause is identical. The putative source clauses of the two *wh*-remnants are identical as well, because of their shared antecedent clause. Below are the types of conjoined *wh*-sluicing that are able to be generated.

- (5) a. Someone ate something, but I don't know **who and what**.
 [*Merger-merger* type]
- b. Sue ate something, but I don't know **what and when**.
 [*Merger-sprouting* type]
- c. Sue ate the meal, but I don't know **how and why**.
 [*Sprouting-sprouting* type] (Adapted from [Citko and Gračanin-Yuksek 2020](#): (2))
- (6) Sue ate something, but I don't know **when and what**.
 [*Sprouting-merger* type] (Adapted from [Citko and Gračanin-Yuksek 2020](#): (2))

As can be seen from the examples in (5), the construction is grammatical regardless of the types of remnants, if all of the argument structure conditions are met by the remnants sharing the same antecedent clause.

It can be noted that the putative source clauses of (5a) are not identical, and partially share some elements in the antecedent, because *who* serves as a subject and *what* as an object. Also, it is questionable whether the *sprouting-merger* type as in (6), is available as well.

2.1.2 Grammatical functions of argument *wh*-remnants

Multiple sluicing has a restricted set of remnant pairs, because of the locality effects that are posed to the construction and the absence of coordinating conjunction '*and*'. For various factors that include those mentioned, they are constrained to either *argument-argument* pair, or *argument-adjunct* pair.

- (7) a. A boy kissed a girl, but I don't know **which boy which girl**. (Adapted from [Kotek and Barros 2018](#): (2c))
- b. I think a boy talked to a girl. I wish I could remember **which boy to which girl**. (Adapted from [Lasnik 2007](#): (66))
- c. I was talking to somebody about something, but I don't recall exactly **to whom about what**. (Adapted from [Richards 2001](#): 139)

In the case of an *argument-argument* pair, as in (7a), the two arguments usually serve different grammatical functions - here, one as a subject and the other as an object. In (7b), while ‘*which boy*’ is a subject, ‘*what girl*’ is a prepositional object. For (7c), both are prepositional objects since they both follow a preposition. However, their grammatical roles are distinct, for ‘*to whom*’ is an argument required by the verb while ‘*about what*’ is an adjunct. In all, multiple sluicing is restricted in the types of *wh*-remnant that are able to undergo sluicing as well as the grammatical functions they serve.

In coordinated multiple *wh*-questions, the types of *wh*-words which are conjoined are constrained in terms of superiority effect, which is a condition that attracts the closest *wh*-word to the attracting head and restricts the movement of other *wh*-words in multiple *wh*-constructions (Sportiche et al., 2013). Since English is not a multiple *wh*-fronting language, the coordination of multiple *wh*-expressions in the Head-initial position can pose syntactic identity issues.

- (8) a. ?* **Who and what** bought?
- b. ?* Who bought and what bought?

- (9) a. Who bought what?
- b. ?? What did who buy? (Grebenyova 2009: (13))

As can be seen in (8a), pairing an *argument-argument* pair in coordinated multiple *wh*-questions can be restricted when the subject is a *wh*-phrase, because finite clause in English requires them to be a definite noun (Citko and Gračanin-Yuksek, 2020). This has been known to be for the EPP requirement (Chomsky, 1981). Even the coordination of the two *wh*-expressions, as in (8b), cannot save the sentence from ungrammaticality.

While leaving one *wh*-word in-situ and moving the other one to the SpecCP position is possible in (9a), it is impossible for any of the *wh*-phrase to violate the superiority effect as in (9b) (Gračanin-Yuksek, 2007; Citko, 2013; Citko and Gračanin-Yuksek, 2020). When there is an overt subject as in (10), coordinated multiple *wh*-question can become felicitous.

- (10) a. **Who and what** did you see?

b. Who did you see?

c. What did you see?

Since the *argument-argument* pair has an overt subject ‘you’ while serving as the same grammatical function, the two *wh*-expressions could be coordinated to undergo a movement operation to the SpecCP position. (10b) and (10c) show that conjoining these clauses are possible for they share the same finite clause.

Conjoined *wh*-sluicing, however, does not need to meet any syntactic requirements in terms of arguments as various types of grammatical functions, which differentiates it from other multiple *wh*-constructions mentioned previously. This may be derived from the property of conjoined *wh*-sluicing that it is, in general, free of locality effects such as clausemate conditions and island constraints, which will be introduced in the following subsections.

(11) a. Some teacher threw something, but I don’t know **which teacher and what**.

(Adapted from Citko and Gračanin-Yuksek 2020: 21) [*Subject, direct object*]

b. Some student has published on some topic, but I forgot **which student and on which topic**. (Adapted from Citko and Gračanin-Yuksek 2020: 20)

[*Subject, prepositional object*]

c. Some student gave some teacher a hard time, but I don’t remember **which student and which teacher**. [*Subject, indirect object*]

Conjoined *wh*-sluicing does not show any preference or restriction as to the kind of grammatical function that it can coordinate, as can be seen in (11). Here, all of the *wh*-remnant pairs are paired with at least one remnant that serves as a subject, which relates to the overt correlate in its antecedent. A question that arises is whether the two *wh*-remnants in the construction can be consisted of the same grammatical functions while sharing a single correlate in the antecedent clause.

2.1.3 SWIPING

Conjoined *wh*-sluicing shares a similar property with multiple sluicing, in not being subject to strict locality restrictions, which also includes **SWIPING** (Sluicing **W**ith **I**nversion of **P**reposition **I**n **N**orthern **G**ermanic) (Browne, 1972; Richards, 2001). SWIPING makes it possible for the sluiced *wh*-remnant with a preposition to be inverted only when the remnant serve as an object.

- (12) I remember Ben was talking with some student about some topic,
- a. ...but I don't know **who with about what**.
 - b. *... but I don't know **who what with about**.
 - c. *... but I don't know **what with who about**.
 - d. *... but I don't know **with who what about**.
 - e. *... but I don't know **who with what about**. (Adapted from Richards 2001: 139)

Postposition is only possible for the preposition of the first *wh*-remnant, but not the second. For conjoined *wh*-sluicing is insensitive to these syntactic conditions, it accommodates more variation of SWIPING cases as in (13).

- (13) I remember Ben was talking with some student about some topic,
- a. ... but I don't know **who with and about what**.
 - b. ... but I don't know **with who and what about**.
 - c. ... but I don't know **who with and what about**. (Adapted from Citko and Gračanin-Yuksek 2020: 11)

As can be seen in (13), the *wh*-remnants with prepositions, whether inverted or not, could be conjoined to form a licit sentence. According to Citko and Gračanin-Yuksek 2020, SWIPING can only occur only if the prepositions and the *wh*-remnants remain relevant in accordance to the argument structure of the antecedent, so (12c) and (12e) should be ungrammatical in cases of conjoined *wh*-sluicing as well because SWIPING cannot cross phrasal boundaries.

2.1.4 Clausemate conditions

Clausemate condition requires the two *wh*-remnants to be derived from a single finite clause (see Takahashi 1994; Nishigauchi 1998; Merchant 2001; Lasnik 2014, and others). Grebenyova 2009 notes that multiple sluicing is acceptable only if the two *wh*-remnants originate in the same clause and follow the ordering sequence of the correlates in the antecedent.

- (14) a. Každýj priglasił kogo-to na tanec, no ja ne pomnju kto
 everyone.NOM invited someone.ACC to dance but I not remember who.NOM
 kogo.
 who.ACC
 ‘Everyone invited someone to dance, but I don’t remember who invited whom
 to dance’
- b. *Každyj priglasił kogo-to na tanec, no ja ne pomnju kogo.
 everyone.NOM invited someone.ACC to dance but I not remember who.ACC
 kto.
 (Russian, Grebenyova 2009: (44))
 who.NOM

It can be observed in both cases of (14), the nominative correlate precedes the accusative correlate and therefore, the ordering of the *wh*-remnants should follow the sequence as well. Here, both *wh*-remnants relates to correlates that reside in the same clause, ‘[*Everyone invited someone to dance.*]’. It is also noted that if the ordering of the *wh*-remnants match that of the correlates in the antecedent, the grammaticality of the construction suffice even when the multiple sluicing construction violates the superiority effect.

Whether the ordering of the two *wh*-remnant is crucial in determining the licit form of conjoined *wh*-sluicing is not determined, nor have been discussed in the previous works. However, it is eminent that this locality effect is not applied to conjoined *wh*-sluicing (Citko and Gračanin-Yuksek, 2020).

- (15) a. *[Some zoologist was disappointed [for Joanne spoke to some economist]], but
 Jim doesn’t know **which zoologist to which economist**.

- b. [Some zoologist was disappointed [because Joanne spoke to some economist]],
but Jim doesn't know **which zoologist and to which economist**. (Adapted
from [Abels and Dayal 2017a](#): 25)

Comparing (15a), a multiple sluicing case with (15b), which is a conjoined *wh*-sluicing one, the difference exemplifies that the latter does not comply to the clausemate restriction. The *wh*-remnants from the construction could be extracted from respective clauses in the antecedent. Yet, it is possible for them to originate from the same clause as well.

- (16) [Jim said [some zoologist was talking to some economist]], but Joanne doesn't
know **which zoologist and to which economist**. (Adapted from [Abels and Dayal 2017a](#): 25)

From (16), '*which zoologist*', which correlates to '*some zoologist*' and '*to which economist*' to '*to some economist*' are both required by the verb '*talk*', which indicates that they reside in the same clause. The requisite in order for multiple correlates to undergo the conjoined *wh*-sluicing operation is not a matter of distance, nor clause bound.

2.1.5 Island constraints

An island violation occurs when a sentence becomes ungrammatical or unacceptable after extracting an element out of a certain phrase or a clause. An element existing inside the island in the antecedent cannot form sprouting cases of sluicing ([Chung et al., 1995](#); [Romero, 1998](#); [Anderbois, 2014](#)).

- (17) a. *Tony sent Mo a picture that he painted, but it's not clear **with what**.
[Complex NP Island]
- b. *Agnes arrived after John ate, but it's not clear **what**. ([Romero 1998](#): (89-90))
[Adjunct island]

However, if that is not the case, the sluicing construction in general is known to repair island violations ([Chung et al. 1995](#), [Merchant 2001](#), [Merchant 2003](#) and many others).

- (18) a. They want to appoint a person who speaks an English dialect, but I don't remember **which**. [Complex NP Constraint]
- b. She will be upset if he talks to one of the girls, but he couldn't remember **who**. [Adjunct island] (Adapted from [Merchant 2003](#): 6)

According to an experimental study based on the acceptability rating, native speakers of English do not distinguish the grammaticality of islands and non-islands if the two remnants are derived from the same clause ([Rodriguez, 2022](#)).

- (19) a. James recalled [that everyone was distracted by something], but he can't remember exactly **who by what**. [*clausemate, non-island*]
- b. James was angry [because everyone was distracted by something], but he can't remember exactly **who by what**. [*clausemate, island*]
- c. [Everyone recalled [that James was hit by something]], but I can't remember exactly **who by what**. [*non-clausemate, non-island*]
- d. [Everyone was angry [because James was hit by something]], but I can't remember exactly **who by what**. [*non-clausemate, island*] (Adapted from [Rodriguez 2022](#): (18))

Non-islands such as *that*-clause ((19a) and (19c)) and strong islands such as *because*-islands ((19b) and (19d)) displayed no statistical significance (a *p*-value higher than that of 0.05) regarding their acceptability rating. However, their acceptability was improved when the two *wh*-remnants derived from the same clause. [Rodriguez 2022](#) explains this phenomenon by suggesting that it is not the island constraint itself, but it is the type of island constraint that can be repaired by the sluicing elliptical construction.

For conjoined *wh*-sluicing, it is possible to predict that the clausemate condition will not exert much influence on the acceptability of the construction unlike in multiple sluicing. Because the construction is not restricted by the clausemate condition, conjoined *wh*-sluicing could be assumed to not being restricted in repairing certain types of island constraints:

- (20) a. James recalled [that someone was distracted by something], but he can't remember exactly **who and by what**.
- b. James was angry [because someone was distracted by something], but he can't remember exactly **who and by what**.
- c. [Someone recalled [that James was hit by something]], but I can't remember exactly **who and by what**.
- d. [Someone was angry [because James was hit by something]], but I can't remember exactly **who and by what**. (Adapted from [Rodriguez 2022: \(18\)](#))

The interpretation of conjoined *wh*-sluicing will be considered in the following section, where it probes into the differences in semantic aspects between other multiple *wh*-constructions with conjoined *wh*-sluicing.

2.2 Semantic properties

2.2.1 Semantics of conjoined *wh*-sluicing

In multiple sluicing, the correlates can hold either a universal quantifier 'every' or an existential quantifier 'some', but not both. But there is a restriction that the universal interpretation should always hold a wider scope than that of the existential one ([Takahashi, 1994](#); [Nishigauchi, 1998](#); [Petesky, 2000](#); [Grebenyova, 2009](#); [Vicente, 2019](#); [Abels and Dayal, 2017a](#); [Kotek and Barros, 2018](#)).

- (21) a. Every boy likes some girl, but I don't know **which boy which girl**. ($\forall > \exists$)
- b. *Some boy likes every girl, but I don't know **which boy which girl**. ($*\exists > \forall$)

As can be seen in (21a), the multiple sluicing construction is felicitous when the universal quantifier takes a wider scope, but when the existential reading is wider as in (21b), the construction become illicit. It should be noted that even though the clausemate condition, or the obligation

for correlates of the remnants to be clause-bounded is satisfied and the ordering of the remnant is matches that of the correlate, following the superiority effect, the sentences become unacceptable due to the scope readings of the two correlates in (21b) (Petesky, 2000; Abels and Dayal, 2017a; Kotek and Barros, 2018).

Merchant 2001 proposes a focus-theoretic implementation regarding the semantic identity of elliptical constructions, which is the conditions given below.

(22) *Focus Condition on Ellipsis (FCE)*

- a. A constituent E can be deleted iff E is e-GIVEN
- b. An expression is considered e-GIVEN iff E has an overt antecedent A and, modulo $\exists$ -type shifting,
 - i. A entails the Focus closure of E (F-clo(E)), and
 - ii. E entails F-clo(A)
- c. F-clo(α) is resulted from replacing F-marked parts of α with $\exists$ -bound variables.

Here, Merchant's FCE analysis can intuitively explain how the Single-Pair reading results from multiple sluicing.

- (23) a. $[\text{TP}_A \text{ Some boy likes some girl}], \text{ but I don't know which boy}_i \text{ which girl}_j [\text{TP}_E t_i \text{ likes } t_j]$.
- b. $\text{TP}_A = \text{F-clo}(\text{TP}_A) = \exists x \exists y [\text{boy}(x) \ \& \ \text{girl}(y) \ \& \ \text{likes}(x, y)]$
- c. $\text{TP}_E = \text{F-clo}(\text{TP}_E) = \exists x \exists y [\text{boy}(x) \ \& \ \text{girl}(y) \ \& \ \text{likes}(x, y)]$ (Kotek and Barros 2018: (25))

Even though Merchant's FCE can explain the existence of at least one entity correlated to the *wh*-remnant, it is insufficient to account for the Pair-List reading, which is a set of answers elicited by the Q+ feature of sluicing. Because the F-clo(E) can only replace α with $\exists$ -bound variables regardless of the scope denoted within F-clo(A), F-clo(E) can only be consisted of existential

interpretations of the *wh*-remnants. So this analysis can be understood to be undergenerating in explaining cases of multiple sluicing.

- (24) Some boy likes every girl, ($\exists > \forall$)
- a. *...but I don't know **which girl which boy**.
- b. ...but I don't know **which girl which boy likes**. (Kotek and Barros 2018: (46))

Here, while it is possible for the existential reading to precede the universal one, the following construction should not be that of a sluicing, as in (24a). The reading can only suffice in an multiple *wh*-interrogative clause in (24b), where only a Pair-List interpretation can arise, asking for boy-girl pairs, as the 'girl' serving as a sorting key in the '*like*' relation (Kotek and Barros, 2018).

There is no "sorting key" for the interpretation in conjoined *wh*-sluicing, because a bi-clausal structure where each remnant forms a respective CP is assumed as its internal structure.

- (25) a. *Every boy likes every girl, but I don't know **which boy and which girl**.
- b. # Every boy likes some girl, but I don't know **which boy and which girl**.
- c. # Some boy likes every girl, but I don't know **which boy and which girl**.
- d. Some boy likes some girl, but I don't know **which boy and which girl**.

It would be possible to apply Merchant's FCE in the case of conjoined *wh*-sluicing, as in (25d). In accordance with this reading, only a single pair of answer could derive from this process, therefore forming a one-to-one relationship rather than one-to-many or many-to-one.

2.2.2 Single-Pair and Pair-List answers

The semantics of conjoined *wh*-sluicing mainly focuses on its reading in terms of Single-Pair or Pair-List answers which are answers to the interrogation raised by multiple *wh*-remnants in the construction. This is related to the quantifier scope of the preceding correlates to their *wh*-remnants (Merchant, 2003; Abels and Dayal, 2017a; Kotek and Barros, 2018; Citko and Gračanin-Yuksek, 2020).

- (26) a. Some student has published on some topic, but I couldn't tell you **which student on which topic**.
[Single-Pair Reading]
- b. Every student has published on some topic, but I couldn't tell you **which student on which topic**. ($\forall > \exists$) [Pair-List Reading]
(Abels and Dayal 2017a: 24)
- (27) a. Some student has published on some topic, but I couldn't tell you **which student and on which topic**.
[Single-Pair Reading]
- b. # Every student has published on some topic, but I couldn't tell you **which student and on which topic**. ($\# \forall > \exists$) [Pair-List Reading]
(Abels and Dayal 2017a: 24)

While it is possible for multiple sluicing to have both readings, the correlates must have the same existential scope as in (26a), or the widest scope must be that of universal quantifier, such as in (26b) (Kotek and Barros, 2018). However, having a universal reading as the wider scope for the remnant is impossible for conjoined *wh*-sluicing in (27b), and is only compatible with existential scope for both remnants as in (27a) (Citko and Gračanin-Yuksek, 2020). This preference in Single-Pair reading also appears in coordinated multiple *wh*-question, which is also conjoined with a coordinating conjunction ‘*and*’ (Citko, 2013).

- (28) **On which topic and when** did you publish?

- (29) Some student has published on some topic, but I couldn't tell you **which student and on which topic**.

Single-Pair reading: Jane has published on the stripping construction.

Pair-List reading: # Jane has published on the stripping construction and Kim has published on the VPE construction. (Abels and Dayal 2017a: 24)

The two constructions, coordinated multiple *wh*-question in (28) and conjoined *wh*-sluicing in (29) have the two *wh*-phrases in the same clause and do not generate any kind of island violation - yet they only yield a Single-Pair reading. The interpretation of both constructions may imply that the two *wh*-phrases in each construction do not form a single clause, but exist in separate CPs that is conjoined by the coordinating conjunction ‘*and*’, producing a single answer from each clause. In the following section, several possible internal structures of conjoined *wh*-sluicing are listed to show how they can induce a Single-Pair reading.

Chapter 3

Previous analyses

The previous analyses in this section are based on the Minimalist framework, where the syntactic structures of conjoined *wh*-sluicing undergo the process of *wh*-fronting, namely the move-and-delete operation. Because there are two remnants, it is impossible for both of the remnants to occupy the SpecCP position. The position of the second *wh*-remnant after the first remnant has gone through *wh*-fronting is controversial as well, in determining if a multiple *wh*-movement is possible. I will introduce four analyses, which is largely divided into three: mono-clause, reverse sluicing, and bi-clausal analyses. The internal structure of the construction is directly related to the readings it can produce, so the structures of multiple *wh*-constructions vary across languages (Chung et al., 1995; Giannakidou and Merchant, 1998; Ginzburg and Sag, 2000; Merchant, 2001; Gračanin-Yuksek, 2007; Citko, 2013; Lasnik, 2014; Kotek and Barros, 2018; Citko and Gračanin-Yuksek, 2020)

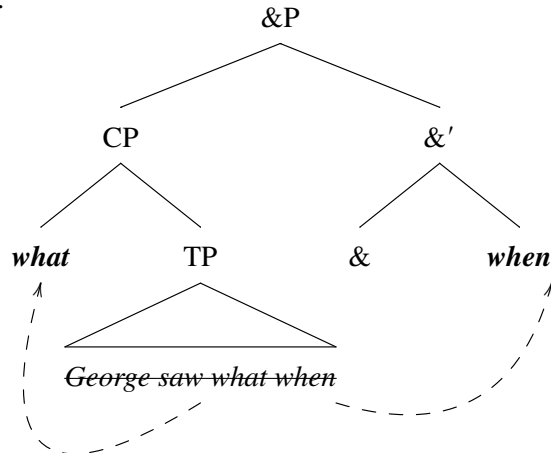
3.1 Mono-clause analysis

The mono-clausal structure requires both of the remnants to form a complex &P and the putative source clause is processed all at once (Zhang 2007, Haida 2007, Haida and Repp 2011, among others). In this way, the remnants would not violate the clausemate condition because they would originate from the same sentence or phrase.

There are two types of mono-clausal structure, the former one is where the first remnant moves to the SpecCP position while the second undergoes rightward extraposition, for English is a non-multiple *wh*-fronting language (Lasnik, 2007, 2014). Here, the remnants exist in different domains, but originate from the same TP. However, it is questionable as to whether the two remnants could share the putative source clause, as can be seen in (2).

- (1) a. George saw something, but I don't know **what and when**.
 b. ...but I don't know [**what_i**] [he saw t_i t_j] [and **when_j**].

- (2) George saw something,
 but I don't know ...



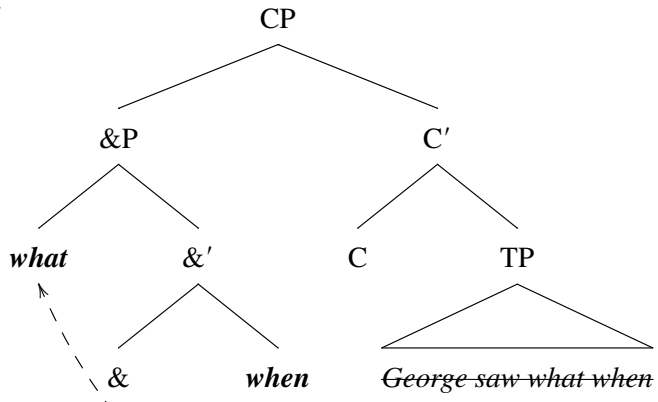
Adapted from (Citko 2013: 297)

Another type of mono-clausal structure is referred to as multiple *wh*-fronting, or sideward movement (Park and Kang, 2007; Richards, 2010), where the two remnants which originate from the same clause both move leftward, but they are not linear for they exist in different domains, as can be seen in (4).

- (3) a. George saw something, but I don't know **what and when**.
 b. ...but I don't know [**what_i**] [and **when_j**] [he saw t_i t_j].

- (4) George saw something,

but I don't know...



Adapted from (Citko 2013: 297)

Only the first remnant can undergo sideward movement up to the SpecCP position here as well, while the second remnant remains inside the &'. It is difficult for this mono-clausal structures to account for conjoined *wh*-sluicing cases when the two *wh*-remnants violate the superiority effect.

- (5) a. I knew I had to do something. I had no idea [[*why*] [*and what*]]. (COCA 1997 FIC)

b. ? ... no idea [*why*_j] [I had to do *t*_i *t*_j] [*and what*_i]

c. ? ...no idea [[*why*_j] [*and what*_i]] [I had to do *t*_i *t*_j]

Comparing (1b) and (5b), the two traces cross paths with each other when the first remnant undergoes *wh*-fronting and the second is extraposed. Here, the superiority effect is violated since the movement to a head is attracts the closest *wh*-remnant, which is '*what*' in (5), not '*why*'. The same logic applies to (5c) which violates the superiority condition when compared with the LF in (3b). As mentioned, the closest *wh*-remnant, '*what*' should move to the Spec position of the &P, but since '*why*' occupies the position instead of the intended remnant. Therefore, the mono-clause analyses are problematic with regards to authentic examples such as (5a).

3.2 Reverse sluicing analysis

From this reverse sluicing analysis by [Giannakidou and Merchant 1998](#), the construction is strictly constrained only to the adjunct denoting remnants, and is unable to host arguments or other complements such as PPs. Also, there can be speaker variances, but most native speakers of English do not systematically distinguish the use of ‘*if*’ or ‘*whether*’.

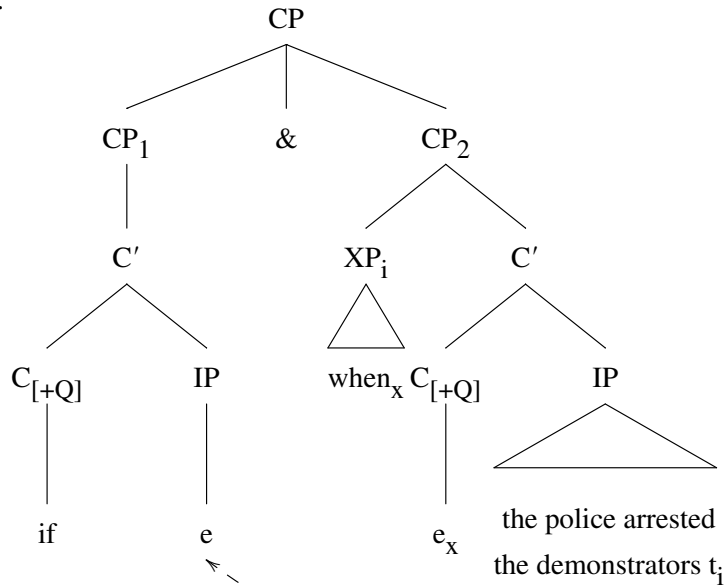
- (6) a. *Lucy was wondering **whether and who** might come to her party.
- b. *The reporters asked **if and who** the FBI had arrested. ([Giannakidou and Merchant 1998](#): (20))
- (7) a. *We found out **whether and on what materials** the company relied.
- b. *Tony’s mother asked **if and how (well)** he had behaved. ([Giannakidou and Merchant 1998](#): (21))

The structure of this construction looks similar to that of the mono-clause structure in that the interpretation of the two remnants originates from the same IP. The difference in this analysis and the mono-clause structure is that the two remnants are separated in distinct CPs which are connected through a ternary structure, conjoined by &.

- (8) a. It’s not clear...
- b. 
[if_i and [when_j the police arrested the demonstrators t_i t_j].

- (9) a. It’s not clear **if and when [the police arrested the demonstrators]**.

b. It's not clear...



(Adapted from [Giannakidou and Merchant 1998](#): 235)

The LF structure in (8) and the internal structure displayed in (9) show that the two CPs complement each other. First, the contextual information of the second CP is shared through the IP recycling, or IP copy process ([Chung et al., 1995](#)). By supplying the empty category in the first remnant, the *wh*-remnant in the latter position acquires a SpecCP position while ‘*if*’ can bear a Q+ reading ([Chung et al., 1995](#)). Another thing to note is that the *wh*-remnant in the second CP can also have a Q+ reading since it is coindexed to the complementizer position. The LF in (8) shows that the IP of the second remnant is copied to the empty complement of the C’ in the first. While the IP is deleted in the clause of the first remnant, it remains overt in the second one, vice versa, while the C[Q+] position is occupied in the first CP, it is left empty in the second, for the *wh*-remnant is not a complementizer and moves up to the SpecCP position. Therefore, the preceding and the latter CPs can both acquire an interrogative clause interpretation, with the second as a *wh*-remnant gaining a full interrogative clause interpretation.

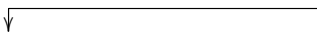
Even though this reverse sluicing analysis can explain how the construction gains an interrogative interpretation and that the two remnants can be positioned at SpecCP in their respective CPs, this analysis fails to account for instances of conjoined *wh*-sluicing where the remnants host

arguments in their remnants.

Also, the reverse sluicing analysis holds a cataphoric dependency of the first remnant on the second. Therefore, the two *wh*-remnants should share the same putative source clause. However, this may be hard to apply to instances of conjoined *wh*-sluicing below:

- (10) a. He decided to use it on someone and it's up to us to find out **who and when**.

(TV 2965 US/CA)

- b. *... it's up to us to find out **who_i** e and **when**[he decided to use it on someone_i].
- 

The correlate of '*who*' is present in the IP of the second remnant, which makes the structure illicit, for there is no gap for the first *wh*-remnant '*who*' to fill - therefore, the reverse sluicing analysis cannot account for cases of conjoined *wh*-sluicing such as in (10a).

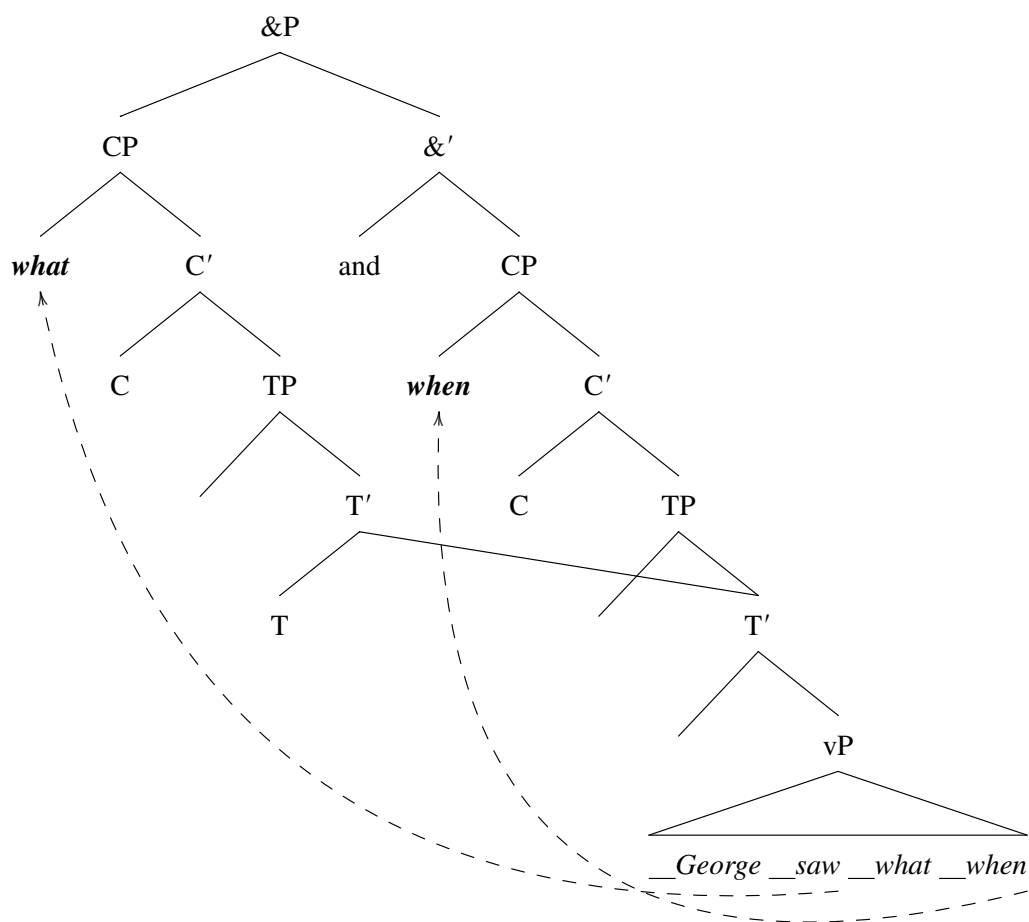
3.3 Bi-clausal bulk sharing analysis

This bi-clausal bulk sharing analysis is somewhat similar to the reverse sluicing analysis, for the two remnants occur in separate CPs. However, they are different in that the reverse sluicing analysis conjoins the two *wh*-remnants in a single CP in a ternary structure, but the bi-clausal bulk sharing analysis connects the two into a single functional &P, which can be observed in the LF below (Gračanin-Yuksek, 2007; Citko, 2013; Citko and Gračanin-Yuksek, 2020).

- (11) a. George saw something, but I don't know **what and when**.

- b. ...but I don't know [**what_i** George saw] and [**when_j** George saw t_i t_j].
- 

- (12) George saw something, but I don't know...



(Adapted from Citko 2013: 310)

Here, the whole vP with both traces of *wh*-remnants is shared between the two conjuncts, with the first remnant sharing all elements within the T' of the second remnant, moving them out to the shared structure. This structure shows dominance of the first remnant on the second one because the whole vP of the second remnant is shared to the first.

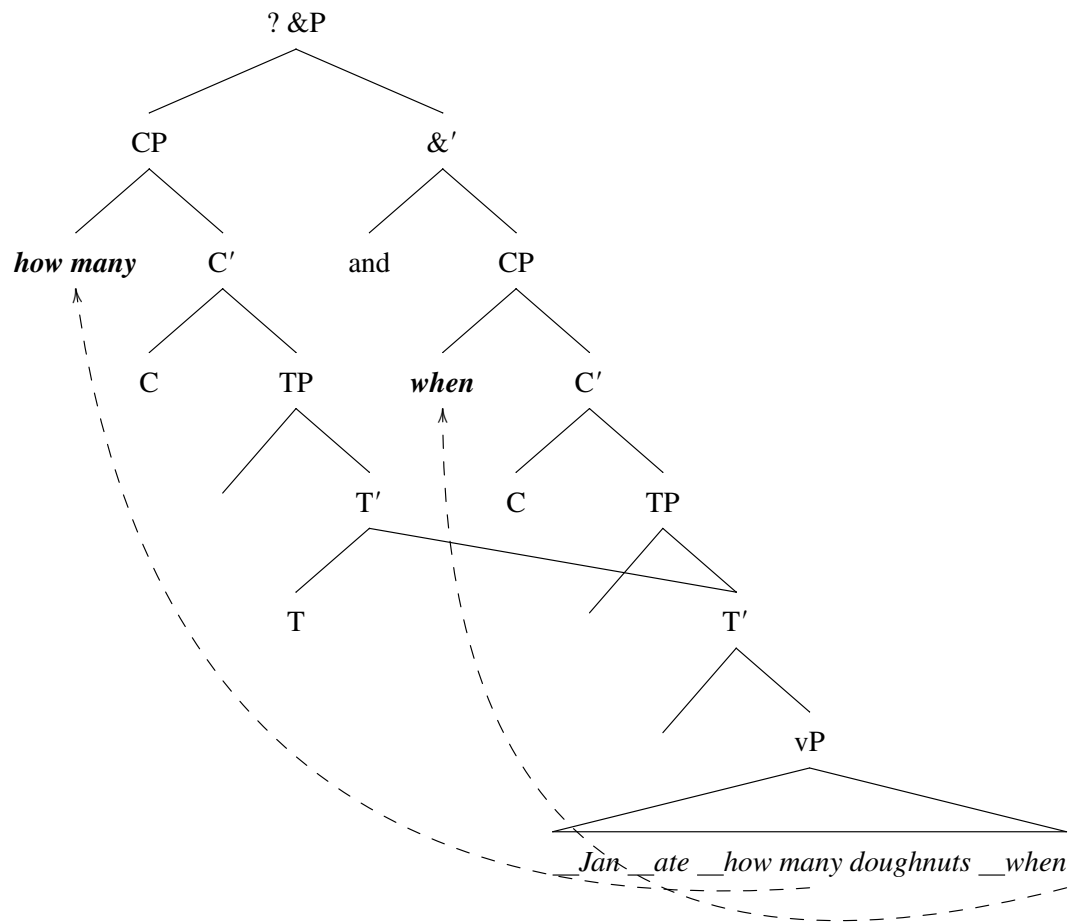
This structure can account for the internal structures of various *wh*-remnant pairs and can conjoin the two remnants to share a putative source clause which exists in the latter remnant, but hold the dominance relationship between the preceding remnant and the latter. This dominance refers to the fact that every element inside the vP should consist the putative source clause of the first *wh*-remnant.

This bi-clausal bulk-sharing analysis is restricted when the *wh*-remnant is a left-branched

quantifier and the structure inside the vP and the *wh*-remnants mismatch (Citko, 2013).

(13) a. Jan ate some doughnuts, but I don't know **how many and when**.

b.



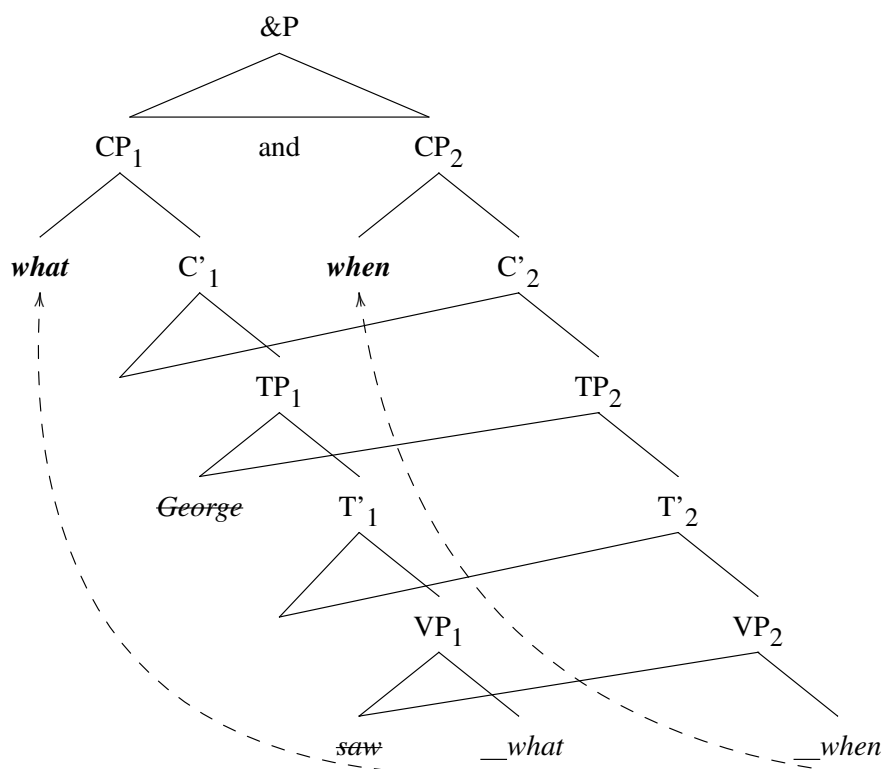
(Adapted from Citko 2013: 322)

The remnant 'how many' is a left-branched quantifier which should appear with the noun 'doughnuts', but does not. Therefore, the putative source clause in the vP of the second remnant does not match to that of the remnant 'how many', making the dominance relation between the *wh*-remnants illicit.

3.4 Bi-clausal non-bulk sharing analysis

The dominance relationship of the preceding bulk sharing analysis is obliterated in the bi-clausal non-bulk sharing, multidominance structural analysis. Here, no remnant exerts dominance over another and therefore has an equal status (Citko, 2013; Citko and Gračanin-Yuksek, 2020).

- (14) George saw something, but I don't know...



(Adapted from Citko 2013: 309)

Based on this structure, Citko and Gračanin-Yuksek 2020 argues that TP₁ and TP₂ are elided for the sake of economy, and the remnants are able to select whichever element they require in generating their respective putative source clauses. While this structure removes the dominance relationship between the two remnants and makes the process of producing putative source clauses more flexible in comparison to any other analysis, this structure cannot account for the case noted

in the bulk sharing analysis, nor when the antecedent alone is insufficient to generate putative source clauses for the remnants:

- (15) “I got a message on my phone a while back and I couldn’t figure out **who and why**.”
(NOW 2020 US)

Even if the contextual cues provide us with sufficient information to derive a putative source clause for the remnants, the antecedent itself is insufficient to serve as a putative source clause for the first remnant *what*, in the case of (15), and only satisfies that of the second remnant ‘*why*’.

In all, the Minimalist framework provides three types of analyses that can account for the internal syntactic structure of the conjoined *wh*-sluicing phenomenon, showing how each remnant is able to generate their putative clause sources through structure sharing or the movement process. However, there are shortcomings within all of the analyses given in this section. Therefore, I believe that an additional investigation is required to observe the distinct properties of this construction to derive a novel account to explain its marked qualities, which will follow in the next section.

Chapter 4

Corpora analyses

4.1 Methodology and data

This thesis investigates the authentic use of the conjoined *wh*-sluicing construction through multiple sources of corpora, including COCA (Corpus of Contemporary American English), BNC, Strathy, NOW (Newspapers on Web), TV, and Movies. The former three were used to collect data from various dialects of English, since the conjoined *wh*-sluicing is not only confined to the American English dialect. The latter three served to balance out the sources of registers, for NOW is a written form of data and the latter two are spoken. Since NOW, TV, and Movies include data from more than six countries (United States, Canada, Great Britain, Ireland, Australia, New Zealand), this corpora investigation strives to accumulate the relevant data to the extent to the dialects of British and Australian English, in order to draw a more generalized observation of the construction.

Below are the search strings implemented in the investigation, which include remnants with prepositions and ‘which + noun’, such as ‘which one’. It should be noted, however, that there was a limitation of five slots that could consist a search string in the BNC corpora. Also, I have narrowed the use of coordinating conjunction only to ‘*and*’, because the use of ‘*or*’ could lead to distinct semantic interpretation relating to a single *wh*-remnant, and it is not a concern of this thesis to determine whether the coordinating conjunction ‘*or*’ would acquire an exclusive or an

inclusive semantic meaning in the conjoined *wh*-sluicing construction.

- (1)
- a. * *wh** and *wh** PUNC
 - b. * *how* and *wh** PUNC
 - c. * *wh** and *how* PUNC
 - d. * *which* NOUN and *wh** PUNC
 - e. * *wh** and *which* NOUN PUNC
 - f. * *which* NOUN and *how* PUNC
 - g. * *how* and *which* NOUN PUNC
 - h. PREP *wh**|*how* and PREP *wh**|*how*
 - i. *wh**|*how* PREP and PREP *wh**|*how*
 - j. PREP *wh**|*how* and *wh**|*how* PREP
 - k. *wh**|*how* PREP and *wh**|*how* PREP
 - l. *wh**|*how* PREP and *wh**
 - m. *wh** PREP and *wh**|*how*
 - n. PREP *wh**|*how* and *wh**
 - o. PREP *wh** and *wh**|*how*
 - p. *wh**|*how* and PREP *wh**
 - q. *wh** and PREP *wh**|*how*
 - r. *wh**|*how* and *wh** PREP
 - s. *wh** and *wh**|*how* PREP

Upon a total of 5,211 tokens searched from the search strings in (1), they were manually sorted out to make up a relevant dataset of 546 tokens. Below is a table that specifies the type of corpora used and how many tokens they each hold.

	COCA	NOW	Movies	TV	BNC	Strathy	Total
frequency	114	261	40	102	21	8	546

Table 4.1: Total number of data collected in each corpus

In the process of sorting the data, irrelevant data was excluded from the dataset. Some of the prototypical examples of impertinent data are those given below:

- (2) a. “They’re always pretty straightforward: Names, dates, **who did what to who and why.**” (COCA 2012 TV)
- b. “That takeaway from the movie stint led you, somewhere along the lines, to run for a seat on the Whanganui council, and even run for mayor. Tell me your **why, how, what and where.**” (NOW 2023 NZ)

The remnants presented in (2a) are multiple *wh*-questions which are placed in-situ, and also used a kind of pronoun, as to denote “someone did something to someone,” rather than to raise an interrogation with the use of *wh*-words. The *wh*-expressions in (2b) as well, are used to signify that the information which can be denoted by each *wh*-word is required. In these examples, it can be seen that *wh*-expressions serve various roles in many different constructions, which required me to scrutinize the data and sort them out to either relevant or irrelevant data.

In the following section, I will elaborate on the four types of variables that I have classified the data into, which will aid the understanding of the properties of conjoined *wh*-sluicing. The variables are in relation to the basic properties of the construction observed in the previous section, namely section 2. The *wh*-remnant pair variable displays an overview of the collected data. Correlate overtness and antecedent clause sharing variables are in relation to the argument structure restriction property, and the grammatical functions of argument denoting *wh*-remnants are observed in corpora data as well.

4.2 Findings

4.2.1 Remnant pairs

The collected data was sorted into four types based on the types of *wh*-remnant pairs they hold: *argument-argument*, *argument-adjunct*, *adjunct-argument*, and *adjunct-adjunct* pairs. Below are the canonical examples of the remnant pairs.

- (3) a. “I feel like somebody should be doing something – but I don’t know **who and what**.” (NOW 2022 AU) [argument-argument pair]
- b. “Something went wrong for him when I involuntarily... sublet my brain. I don’t know **what and why**.” (COCA 2004 FIC) [argument-adjunct pair]
- c. “The only place they could be was in one of the island’s abandoned artillery bunkers. The question was **where and which one**.” (TV 1987 US/CA) [adjunct-argument pair]
- d. “They’re going to call back in a few hours, let us know **where and when**.” (TV 1992 UK/IE) [adjunct-adjunct pair]

The data holds not only just canonical *wh*-remnant types, but noncanonical ones as well. In (3), the *adjunct-argument* pair, which was not mentioned in the previous literature, appears in the dataset. This ordering of the *wh*-remnants might show difference from the much more canonical *argument-adjunct* pair with respect to the information packaging of the context - the argument *wh*-remnant *which one* in (3c) could have been placed at the end of the sentence to receive focus. Because (3b) and (3c) is paired with an adjunct, further analysis should be conducted to determine whether this ordering of the *wh*-remnants have differences. Below is a pie chart showing the distribution of the *wh*-remnant pair types collected in this thesis.

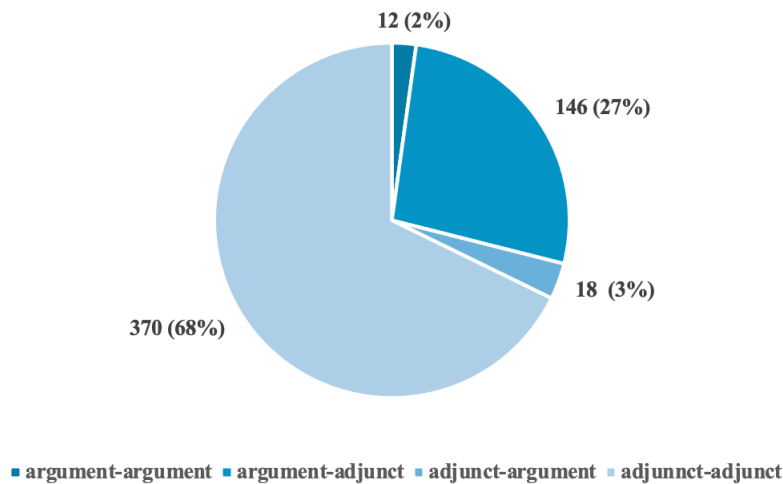


Figure 4.1: Distribution of *wh*-remnant pair types

First, it can be observed that the conjoined *wh*-sluicing construction prefers the *adjunct-adjunct* pair the most, for this *wh*-remnant pair usually do not require an overt correlate and can be classified as a sprouting type of the elliptical construction. The least favored is the *argument-argument* pair, and it is noticeable that even though the grammaticality of the pair is enhanced with coordination, it is still unfavored.

Another observation is that the *argument-adjunct* pair is more preferred than the *adjunct-argument* pair. This preference of the ordering of the *wh*-remnants is parallel to the that in the information structure, because familiar information appears before the new information. Since the overt correlate of the argument appears in the antecedent, it could be predicted that the speakers would prefer to present the argument-denoting *wh*-remnant earlier than the adjunct-denoting remnant, hence showing a preference of *argument-adjunct* pair over the *adjunct-argument* pair.

Another thing to note in this variable is that there are adjunct-denoting *wh*-remnants which are not confined to *where*, *when*, *why*, and *how*. There are instances where the remnants hold prepositions as well.

- (4) a. "If you're going to attack the Cybermen with missiles, they're going to retaliate, and we want to know **how and with what.**" (TV 1968 US/CA)
- b. "They must be stopped. The only question is **by whom and when.**"

(NOW 2016 US)

These remnants are classified as *adjuncts* this is because (4a) inquires the manner and the tool that will be used in terms of acting out that manner. In the following example, (4b), the argument structure restriction is satisfied by the remnant *by whom*, since the antecedent structure is in a passive voice and also inquires the time of the action with the following remnant *when*.

4.2.2 Correlate overttness

This variable observed whether the two *wh*-remnants in the construction had overt or covert correlates in the antecedent site. This correlate overttness variable can be interpreted as merger or sprouting type of the *wh*-remnant pairs, which are both hosted by the construction.

- (5) a. “Someone was keeping something a secret. The only way to get Gella out of there is to find out **what and who**.” (COCA 1991 FIC) [overt-overt pair]
- b. “If one of them did get Greely up here, I want to know **which one and why**.” (TV 2007 UK/IE) [overt-covert pair]
- c. “Someone obviously decided to do something about it, and we wonder **why and who**.” (BNC 1985-1994) [covert-overt pair]
- d. “We know Bremen’s gonna make another move. It’s just a matter of **where and when**.” (TV 1982 US/CA) [covert-covert pair]

Examples in (5) are canonical examples regarding overttness of the *wh*-remnant pairs. Indefinite nouns serve as the correlates of the *wh*-remnants, as in (5b). It is easier to generate the sluiced site of the *wh*-remnants when overt correlates are given in the antecedent clause. Below is the table regarding the distribution of this variable.

	COCA	NOW	Movies	TV	BNC	Strathy	Total
overt-overt	5	6	1	2	3	1	18 (3%)
overt-covert	25	15	5	28	10	0	83 (15%)
covert-overt	7	6	2	1	2	0	18 (3%)
covert-covert	77	234	32	71	6	7	427 (79%)
total	114	261	40	102	21	8	546 (100%)

Table 4.2: Overt/Covert pairs

While *overt-overt* cases are rare, *covert-covert* cases are productive. This frequency difference may be due to the property of the conjoined *wh*-sluicing mentioned in the previous subsection that it prefers the *adjunct-adjunct* pair, which are mostly sprouting types. The preference of the *argument-adjunct* pair is reflected in this variable as well, since the raw frequency of *overt-covert* pairs is greater than that of *covert-overt* pairs.

From the pairs given, it can be inferred that argument-denoting *wh*-remnants are the ones that prefer overt correlates while the adjunct-denoting ones do not. However, that is not always the case. There are instances where the adjunct-denoting *wh*-remnants can be seen to have overt correlates in the antecedent as well.

- (6) “We’re all gonna die sometime. It’s just a question of **how and when**.” (Movies 1994 US/CA)

In (6), there is a correlate ‘*sometime*’ for the remnant *when*. In the other respect, it is possible for argument-denoting *wh*-remnants to have covert correlates, much less a clear antecedent source clause at all.

- (7) a. “Are they going to tell people?” She nodded vigorously. “Absolutely. The problem is **how and what**.” (NOW 2020 US)
- b. His lower lid almost swelled the whole left. eye closed. I taped his right eye shut. I said, “This will help,” but didn’t say **who and what**. (COCA 2001 FIC)

The first example (7a) implies that there is something to be told to people, but does not exactly overt in the antecedent. However, there is some relevant context to refer to. On the other hand, the remnants in (7b) do not have a clear antecedent to refer back to. For example, the reconstructed form could be ‘who this will help’ or ‘who I am’, and for *what*, it could be ‘what this is’ or ‘what will help him’, and so on. In order to accurately interpret the sluiced site, we must take an extensive account to the context.

4.2.3 Antecedent clause sharing

This variable observed the distribution of the two *wh*-remnants sharing the identical putative source clause derived from the antecedent. It is important that while the *completely* and *partially* sharing types are able to refer to the antecedent clause to produce their putative source clause, but this is impossible for the *not sharing* type, which is when the two *wh*-remnants nor the antecedent share the exact putative source clause.

- (8) a. Empton knew about O’Shaughnessy long before we did. I want to know **how**
 <Empton knew about O’Shaughnessy long before we did>
 and **why** <Empton knew about O’Shaughnessy long before we did>.
 (TV 2008 UK/IE) [Completely sharing]
- b. Something had been achieved, even though he was not quite sure **how** <—_i
 had been achieved> and **what**_i <*had been achieved*>. (BNC 1985-1994)
 [Partially sharing]
- c. The football weighs more than he does. Wait. How did... I don’t know **how and**
 what. (COCA 2019 MOV) [Not sharing]

The *completely sharing* type of (8a) has both of the *wh*-remnants share the identical putative source clause, which is directly derived from the antecedent. (8b), which is the *partially sharing* type, shares the putative source clause between the antecedent and the remnants, but they are slightly distinct between the two *wh*-remnants. The index of the second remnant ‘*what*’ is required to satisfy the argument structure requirement of the preceding remnant *how*, which is reflected in

its putative source clause. Lastly, (8c), the *not sharing* type has neither the antecedent nor the putative clause sources shared with one another.

Below is a distributional table on the sharing types within the data.

	COCA	NOW	Movies	TV	BNC	Strathy	Total
completely sharing	68	197	22	60	12	4	363 (66%)
partially sharing	34	48	16	33	6	2	139 (26%)
not sharing	12	16	2	9	3	2	44 (8%)
total	114	261	40	102	21	8	546

Table 4.3: Antecedent putative source cases

As can be seen from the table, the *completely sharing* cases are the most frequently occurring, making up two-thirds of the data. However, a third of the data consists of *partially* and *not sharing* types. This distribution of the dataset shows that the analysis of the construction should concern how the putative source clause could be shared between the two *wh*-remnants.

4.2.4 Grammatical functions of argument-denoting *wh*-remnants

The grammatical functions of argument-denoting *wh*-remnants are various and not restricted to subject and object grammatical functions. Five grammatical functions were observed within the data, exemplified in the examples below.

- (9) a. He and Newsom have warned that more restrictions could be coming, but they're not saying **what and when**. (NOW 2020 US) [Subject]
- b. You're holding back. I wanna know **what and why**. (TV 2013 US/CA) [Object]
- c. Why would Naharana want the Immobilo? Well, I'm not sure, but I think he plans to use it on someone, and it's up to us to find out **who and when**. (TV 1965 US/CA) [Prepositional object]

- d. She knew this was an important occasion, although she could not remember precisely **what and why**. (COCA 1998 FIC) [Predicative complement]
- e. You need someone's help , but I don't know **whose and why**.
(TV 1990 US/CA) [Possessive]

The subject denoted by the *wh*-remnant by '*what*' in (9a) can be inherited from the antecedent, as in '*what could be coming*', for '*more restrictions*' in the antecedent is the overt correlate for '*what*'. In (9b), the correlate, though not overt, could be interpreted through the putative source clause, as in '*you're holding back (something)*'. The prepositional object in (9c), where the *wh*-remnant '*who*' correlates to '*someone*' in the antecedent, appears after the preposition 'on'. The '*what*' in (9d) serves as a subject predicative complement, for the overt correlate of this *wh*-remnant is '*an important occasion*' which is followed by a linking verb, copula *be*. Lastly, though rare, possessive cases occur in the data as well, as in (9e), in which the putative source clause of '*whose*' would be directly derived from the antecedent as '*whose help you need*'.

Below is a distributional bar graph of how the grammatical functions of argument-denoting *wh*-remnants are dispersed throughout the dataset.

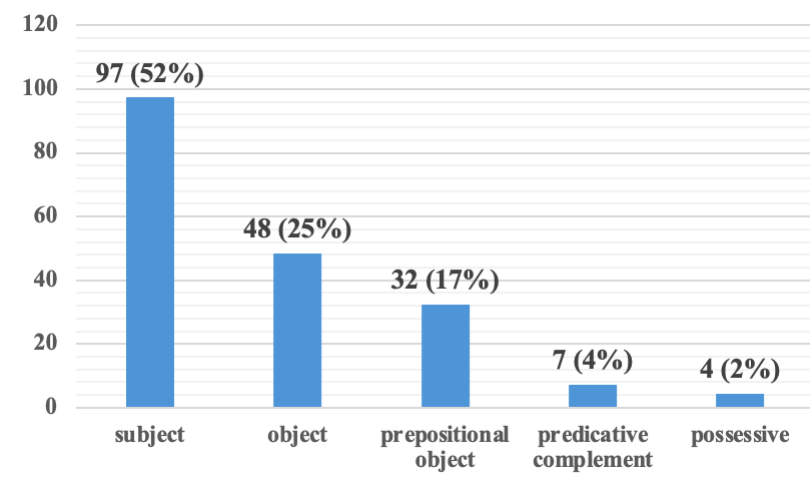


Figure 4.2: Grammatical functions of argument-denoting *wh*-remnants

The figure above shows us that subject denoting *wh*-remnants make up more than half of the data, followed by the object and the prepositional object consist 42% of the data when combined.

On the other hand, the predicative complement and possessive functions are rare, each accounting for less than 5% of the data.

Sometimes, there are cases where the argument-denoting *wh*-remnant share the same correlate.

- (10) “If God had any hand in this, then life was a slaughterhouse. When the sadness erupted over his happy life, the abyss opened beneath him and he fell. In this headlong plunge he instinctively reached out and grabbed hold of something, he didn’t know **who and what**.” (COCA 2011 FIC)

In this case, the overt correlate ‘*something*’ in the antecedent can serve as an overt correlate for both *wh*-remnants. This instance is denoting the ambiguity or the uncertainty of the animacy of the thing/the person he held onto in the given context. However, both *wh*-remnants could be derived from the same clause with different indefinite nouns as their overt correlates.

- (11) a. “Someone was keeping something a secret. The only way to get Gella out of there is to find out **what and who**.” (COCA 1991 FIC)
- b. “I feel like somebody should be doing something – but I don’t know **who and what**.” (NOW 2022 AU)

These instances in (11) are both following the clausemate condition, and they can be seen as being highly productive when inquiring for more than one information. Also, they also display that argument-denoting *wh*-remnants have a preference to have overt correlates in their antecedent, which is related to the correlate overtness variable aforementioned. In the following section, I will extensively observe idiosyncratic instances appearing within the dataset, which raises questions with regards to the existing analyses of conjoined *wh*-sluicing.

4.3 Discussion

4.3.1 Violation of structural parallelism

There are cases where the putative source clauses generate structures that are distinct from those of the antecedent. This distinctiveness could vary in a sense they could be partially distinct to not

being parallel at all. As have been discussed in putative source clause completely sharing case, it is canonical for both of the *wh*-remnants of conjoined *wh*-sluicing to produce the identical putative source clause between them as well as with their antecedent.

- (12) a. “And, if someone somehow hacked into the Fortress of Solitude, then Zoe can help us figure out **who and how**.” (TV 2016 US/CA)
- b. “He told us he met Freddy the Frog but he didn’t say **where and how**.” (TV 1987 UK/IE)

In cases where there are overt correlates in the antecedent or if the *wh*-remnant pair is sprouting, the putative source clauses match with those of the antecedent. However, they can partially match, referencing the antecedent in forming the putative source clause for each remnant.

- (13) a. “I got a message on my phone a while back and I couldn’t figure out **who and why**.” (Repeated from (13) of section 3) (NOW 2020 US)
- b. “No woman is worth a bullet between the eyes. It depends on **whose eyes and which woman**.” (Movies 1991 US/CA)

In (13), the antecedent holds contextual information which involves in reconstructing the putative source clause of both *wh*-remnants, but the antecedent itself cannot serve as well-formed putative source clause. The putative source for ‘*why*’ in (13a) is not a problem, for it is possible for the *wh*-remnant to acquire the antecedent clause itself as its putative source clause. However, that is not the case for the other *wh*-remnant ‘*who*’. Not the antecedent clause itself, but the contextual cues from the antecedent should be implemented to form a well-formed putative source clause for the remnant. In (13b), the remnant ‘*whose eyes*’ refers to the prepositional object of the antecedent sentence and ‘*which woman*’ is indicating the subject. Therefore, a possible reconstructed form for each instances could be:

- (14) a. ...and I couldn’t figure out **who** messaged my phone a while back and **why** I got a message on my phone a while back.

- b. It depends on **whose eyes** a woman is worth a bullet between and **which woman** is worth a bullet between the eyes.

The putative source clauses for both examples in (13) utilizes the contextual and syntactic identity in the antecedent clause to better form a structure for each of the remnants, as in (14). Hence, these *wh*-remnants should modify the preceding antecedent clause to be structurally well-formed for each of its *wh*-remnants. Lastly, there are cases where it is impossible to generate a well-formed putative source clause from only referring to the syntactic structure of the antecedent clause.

- (15) a. His lower lid almost swelled the whole left. eye closed. I taped his right eye shut. I said, "This will help, " but didn't say **who and what**. (COCA 2001 FIC)
- b. "You know, there are times when watching you two swinging your at each other is quite funny, but we've just lost 45 grand and we're still scratching our heads trying to figure out **how, and who**." (TV 2007 UK/IE)

From the instances above, the putative source clauses for both argument-denoting *wh*-remnants in (15a) and '*who*' in (15b) require to be constructed through contextual cues, not directly from the syntactic information given in the antecedent. There is no clear antecedent for the *wh*-remnants to form a putative source clause from. If the antecedent as an IP were to be recycled or shared with the putative source clauses of the *wh*-remnants in both examples, the sentences formed will be illicit.

- (16) a. ? I said, "This will help," but didn't say **who** this will help and **what** this will help.
- b. # ..., but we've just lost 45 grand and we're still scratching our heads trying to figure out **how** we've lost 45 grand and **who** we've lost 45 grand.

Therefore, if the analyses given in section 3 were to be applied to these cases of conjoined *wh*-sluicing of antecedent clause *partially* and *not sharing*, the construction would not be able

to form well-formed putative source clauses for the *wh*-remnants. This observation implies that when constructing a putative source clause, not only the syntactic structure, but also the semantic meaning is required.

4.3.2 SWIPING cases

As mentioned before, SWIPING, a case of inversion with a preposition and its sluiced *wh*-remnant, was observed in the collected dataset. There are instances where both of the *wh*-remnants were with prepositions, but as can be seen (18), the instances where there is only one *wh*-remnant with a preposition but the other does not are more frequently occurring.

- (17) a. "SEPA fails on the most basic question for corporates – when they receive money they want to know **from whom and for what**." (NOW 2016 US)
- b. "The ship did take some time to do a search, I forgot **what for and for how long**." (2022 NOW US)

In (17), the elided putative source clauses for both *wh*-remnants are '*they receive money*' and '*the ship take some time to search*' respectively. It should be noted that only two cases of both *wh*-remnants with prepositions occurred within the database. In (18), instances of conjoined *wh*-sluicing with a single *wh*-remnant with SWIPING appeared with another *wh*-remnant that denotes adjuncts rather than arguments.

- (18) a. "You know things have happened and you don't know **when and to whom**," he says. (2020 NOW CA)
- b. "We were expecting this to happen and we just didn't know **when and to what extent**," she said. (2020 NOW US)
- c. They robbed him and did that to him, for what? I want to know **why and for what**. (2022 NOW UK)
- d. "From a purely scientific perspective it would be interesting to have some cars that are marked as self-driving cars and some that are not and see whether

other road users react in a different way," Volvo technical leader Erik Coelingh told the paper. I would expect they will, but I don't know **how and to what extent**." (2016 NOW Australia)

- e. "They have spent a lifetime with the ball in their hands, and while they are willing to share it, they decide **when and with whom**." (COCA 2014 MAG)
- f. "Yes, when they felt wounded and needy, they'd allow her to wrap her arms around them, demanded it even. But they always decided **when and for how long**." (COCA 2011 FIC)

As can be seen from many instances given in (18) above, cases with a *wh*-remnant with a preposition that does not undergo SWIPING occur in various dialects and context, but mostly in written form of the English language. So, it can be inferred that SWIPING is likely to appear in the construction where both of the *wh*-remnants have prepositions. Also, the examples given above are *adjunct-adjunct* pairs, so it can be generalized that *wh*-remnants with prepositions prefer to be paired with another adjunct-denoting *wh*-remnant. However, there is a case where SWIPING did occur.

- (19) If George drinks, I wanna know **who with and where**. (COCA 2004 TV)

Unlike the cases observed above in (18), the inverted preposition appears in the preceding position of the *wh*-remnant, and appears in spoken register. This can imply that SWIPING occurs in rather informal than formal form of registers and serves as an adjunct-denoting remnant to inquire additional information about the current topic in discussion.

4.3.3 Island violations and the clausemate condition

This corpus investigation reveals that conjoined *wh*-sluicing repairs island violations, just like the sluicing construction with a single *wh*-remnant.

- (20) a. "Are those... fingerprints?" "Yeah, that's what they look like."
"From someone holding him down?" "I don't know **how and who**". (COCA
 2001 TV) [Adjunct island]

- b. *Who is it from ____ holding him down?
- (21) a. “We did not know where he was going and who he was meeting with. If he had relayed that information early on, it may not have been avoided, but it could have been accelerated a lot quicker if we knew **who and where**.” (NOW 2019 US) [Coordinated structure constraint]
- b. *Where didn’t we know he was going ____ and who he was meeting with?
- c. *Who didn’t we know where he was going and he was meeting with ____?

The conjoined *wh*-sluicing construction makes it possible to extract elements from these islands without being ungrammatical, therefore repairing island constraints. Remnant ‘*who*’ in (20) is extracted from an adjunct clause with ‘*from*’, which would form an ungrammatical sentence, but the flow of the discourse is licit due to island repair done by conjoined *wh*-sluicing. Similarly, the two *wh*-remnants could be extracted from a coordinated structure in the antecedent in (21) because the conjoined *wh*-sluicing repairs its ungrammaticality.

Also, the clausemate condition violation can be observed in the dataset but there is a preference for the correlates of the *wh*-remnants to be a clausemate in the antecedent clause.

- (22) a. [Someone was keeping something a secret]. The only way to get Gella out of there is to find out **what and who**. (COCA 1991 FIC)
- b. I feel like [somebody should be doing something] – but I don’t know **who and what**. (NOW 2022 Australia)

In (22a), both of the overt correlates are in the same clause - therefore the *wh*-remnants ‘*what*’ and ‘*who*’ would not violate the clausemate condition. Also, (22b) shows a similar case as well, because ‘*who*’ serves as the subject in the clause while ‘*what*’ functions as the object in the same clause. Although it is rare, the two *wh*-remnants of the construction could violate the clausemate condition.

In addition to these properties that have been observed in the previous works, following are additional instances show that the interpretation of the conjoined *wh*-sluicing is not able to be solely based on its syntactic aspects.

4.3.4 Preposition drop

According to [Sag and Nykiel 2011](#), preposition drop and clefting cases are possible in sluicing, and this applies to conjoined *wh*-sluicing as well.

- (23) a. He was devastated and suggested we stay together but I could sleep with other people, as long as I told him **who and when**. (NOW 2017 IE)
- b. She's interfacing with other cells in the deep Web. ... We assume other like-minded groups. But we don't know **who and where**. (COCA 2013 MOV)

In (23), the two cases show that the preposition 'with' is dropped with the *wh*-remnant '*who*', even though it appears in the antecedent with an overt correlate. Not only the preposition 'with', but also frequent cases are those with 'on' or 'of' as well.

- (24) a. "Well, I'm not sure, but I think he plans to use it on someone and it's up to us to find out **who and when**." (TV 1965 US/CA)
- b. "In this headlong plunge he instinctively reached out and grabbed hold of something, he didn't know **who and what**." (COCA 2011 FIC)
- c. "..., the universe attempts to carry on as if she had died, causing much confusion as the citizen begin mourning the loss of someone though they don't know **who and why**." (COCA 2012 WEB)

From this property, it is possible to infer that the construction of conjoined *wh*-sluicing drops prepositions when they appear with argument-denoting *wh*-remnants. This phenomenon also occurs when there is a voice mismatch between the antecedent and the putative source clause of the *wh*-remnant.

- (25) a. "I think I know how Mrs. Garrity was killed. I think I know **who and why**." (TV 2016 UK/IE)
- b. "So we have a idea of what killed him. We only need **whom and why**." (TV 2003 US)

When the antecedent clause is in a passive voice, the argument-denoting *wh*-remnant that follows is rarely with the preposition ‘by’, as in (25). The *wh*-remnants can be a nominative form as in (25a), or take on the accusative form like in (25b). This preposition drop can also account for the property of the construction that violates the argument structure requirement posed by the antecedent when forming the putative source clauses of the remnants. This property of conjoined *wh*-sluicing that further strengthens its characteristic of insensitivity to the locality constraints.

4.3.5 Possessive *wh*-remnants

The bi-clausal bulk-sharing analysis was problematic due to the left-branching of the quantified remnants. The data analysis revealed that the possessive *wh*-remnant ‘*whose*’ is sluiced while the antecedent implies that there is a correlate which should be linked to the possessive expression.

- (26) a. “Then again, Gates may not care so much how quickly Microsoft’s Internet products appear, as long as the marketplace is expecting them. In his October 1994 staff memo, he noted that when a sea change occurs and customers have no choice but to “upgrade ” to a new way of computing, the only questions are **whose and when.**” (COCA 1995 MAG)
- b. “There will be blood, the only questions are **how much and whose.**” (NOW 2018 CA)

The interpretations of the examples in (26) will depend on the contextual cues, and not directly derived from their antecedent clauses. The possessive case will require a putative source clause with a noun following the case marker such as “*whose Internet products*” in (26a) and “*whose blood*” in (26b).

This property of conjoined *wh*-sluicing also supports the claim made in this thesis that the syntactic property of the construction alone is insufficient to explain or interpret the putative source clauses of each *wh*-remnant. Along with its structural aspects, its semantic interpretation as well as pragmatic properties should be considered as well.

4.3.6 Covarying collexeme analysis

According to [Stefanowitsch and Gries 2005](#), when a construction provides more than two slots of variables which hold semantic properties that is related to one another, a covarying collexeme analysis can be done, in order to find any statistically meaningful semantic relation regarding the frequency of one variable to another.

Even though the analysis is usually implemented to validate the attraction of certain types of verbs to a construction, this study applied this analysis to define the attraction between the two *wh*-remnants in conjoined *wh*-sluicing, regarding their type and ordering. The covarying collexeme analysis was done to observe whether the position of a certain *wh*-remnant influences that of the following *wh*-remnant, and to validate their relationship with statistical figures.

A 2-by-2 table was made, where the first slot consisted of the frequency of the first conjunct as whether ‘argument’ or ‘adjunct’, and the same has been applied for that of the second conjunct. Observe the table below:

	argument _{Slot2}	adjunct _{Slot2}
argument _{Slot1}	12	146
adjunct _{Slot1}	18	370

Table 4.4: A 2-by-2 table of types of *wh*-remnants

With this table, the code for covarying collexemes was implemented from the webpage of [Stefanowitsch and Gries 2005](#), where a Generalized Linear Model was used with specific log values. The outcome of this statistical analysis revealed a table that showed either ‘attraction,’ which means that the two variables had some statistical significance with one another or ‘repulsion’, which shows that they do not. The table¹ below shows which remnant pairs resulted as ‘attraction’.

¹ Note that the LLR values are rounded up to the one-hundredth decimal point.

Attracted pairs	LLR	Repelled pairs	LLR
what-when	79.51	who-how	-0.01
how-why	465.83	who-when	-0.01
when-where	463.72	who-why	-0.09
why-how	255.48	what-why	-1.27
where-when	203.41	who-where	-3.07
what-who	14.02	when-who	-4.70
who-what	21.73	how-who	-4.90
where-who	8.18	when-what	-11.53
where-what	6.80	how-what	-10.45
why-what	6.39	what-why	-10.07
what-where	4.68	what-where	-16.62
what-how	1.01	where-how	-29.22
why-who	0.53	why-when	-29.90
		how-when	-45.17
		where-why	-52.64
		when-how	-71.11
		why-where	89.53
		when-why	-130.27
		how-where	-136.29

Table 4.5: *Wh*-remnant pairs with attraction and repulsion

Some of the inferences that can be given from the table is that first, there are less pairs with attraction than with repulsion. This means that only a few types of remnant pairs are preferred in usage in comparison with possible pairs of remnant. Another point is that while some pairs with different *wh*-remnant values are repelled when reversed, such as ‘what-when (79.510)’ and ‘when-what (-11.53)’. Therefore, this statistical analysis reveals that some remnant pairs are preferred in a certain position with a preferred *wh*-remnant type. Lastly, the types that do not occur at all, such as ‘who-who’, ‘when-when’, or ‘where-where’ is repelled, which means that they are not utilized in the discourse.

Chapter 5

Towards a construction-based analysis

Given some of the aspects which is discussed in the previous section regarding some properties of conjoined *wh*-sluicing, where some of the instances found in the corpus data cannot be accounted for by the analyses from the Minimalist framework, I would like to conjecture a possibility of an approach to conjoined *wh*-sluicing based on the Construction Grammar (CxG) and Head-driven Phrase Structure Grammar (HPSG), which does not solely resort to the syntactic aspect but considers semantic, pragmatic aspects of the construction as well ([Ginzburg and Sag, 2000](#); [Culicover and Jackendoff, 2005](#); [Ginzburg and Fernandez, 2010](#); [Kim, 2013](#); [Kim and Nykiel, 2020](#); [Abeillé and Chavez, 2021](#)).

Conjoined *wh*-sluicing conjoins two respective *wh*-remnants which either denotes entity or proposition with an interrogative force, deriving the [Q+] feature in its internal syntactic structures. In other words, each conjunct in conjoined *wh*-sluicing can be considered as having a propositional meaning or denoting an entity ([Ginzburg and Sag, 2000](#)).

I would like to assume that the interpretation of a single sluiced clause gains its meaning in terms of its ‘question-under-discussion (QUD)’ value in the dialogue, in which QUDs are processed as a continuous sequence of information in the Dialogue Game Board (DGB), which keeps track of what has been said to whom, and which entity refers to who/whom/what ([Ginzburg and Fernandez, 2010](#)). Here, the semantic meaning of nonsentential utterances which can undergo maximal projection can be resolved in terms of the contextual parameters given by the DGB. Because the

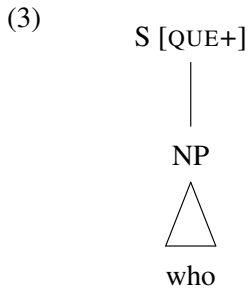
QUDs are constantly being updated as the conversation continues, the interpretation of each sluiced clause in conjoined *wh*-sluicing can be acquired from the QUD values given by the DGB, which requires two attributes in minimum, namely salient utterance (SAL-UTT) and maximal-question-under-discussion (MAX-QUD) (Kim, 2015).

$$(1) \quad \left[\begin{array}{c} \text{DGB} \left[\begin{array}{c} \text{SAL-UTT ...} \\ \text{MAX-QUD ...} \end{array} \right] \end{array} \right]$$

Here, the MAX-QUD is the latest QUD in the discourse and the SAL-UTT represents the syntactic and semantic evidences given in the antecedent. This approach, which supposes that the meanings derived from the unpronounced materials in conjoined *wh*-sluicing are generated without any overt syntactic structure, neither do they rely solely on the syntactic structure in the antecedent. Instead, the single daughter of an S-node, which is referred to as a fragment, is directly projected to its maximal projection (see Ginzburg and Sag 2000 and Culicover and Jackendoff 2005).

(2) **Head-Fragment Construction** (Kim and Nykiel 2020):

Any category can be projected into a NSU (non-sentential utterance) when it functions as a salient utterance (SAL-UTT).



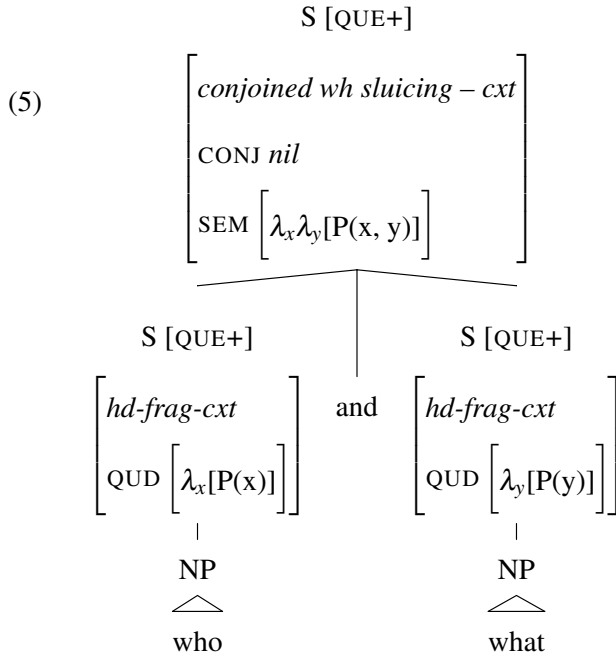
(Kim, 2015)

As given in the tree in (3), each *wh*-remnant would have an internal structure where it would be directly projected into a sentence to acquire a sentential meaning. Other aspects to consider would be how the two *wh*-remnants would be conjoined. The corpus data from the previous Section suggests that the internal structure of the construction would most likely take on a structure which combines the two semantic representations of *wh*-remnants in a larger unit.

According to [Bilb   and Gazdik 2012](#), the coordinating conjunction ‘and’ serves as a functor conjoining two *wh*-phrases. For conjoined *wh*-sluicing, I would like to postulate the same function to the coordinating conjunction in the construction as well.

Regarding the aspects mentioned, I would like to propose a *conjoined wh-slucing* construction where the two *wh*-remnants serve as respective variables of a sentence which is projected into acquiring a sentential meaning through the two *Head-Fragment* construction whose meanings could be combined into a whole sentential meaning. Since the syntactic categories of the *wh*-remnants may differ as adjunct or argument, each remnant should be projected into having a sentential meaning in order to be coordinated.

$$(4) \quad \left[\begin{array}{c} \text{conjoined } wh\text{-sluicing-cxt} \\ \text{CONJ } nil \\ \text{SEM } [\lambda_x \lambda_y [P(x, y)]] \end{array} \right] \rightarrow \left[\begin{array}{c} \text{hd-frag-cxt} \\ \text{QUD } [\lambda_x [P(x)]] \end{array} \right], \left[\begin{array}{c} \text{CONJ } nil \\ \text{hd-frag-cxt} \\ \text{QUD } [\lambda_y [P(y)]] \end{array} \right]$$

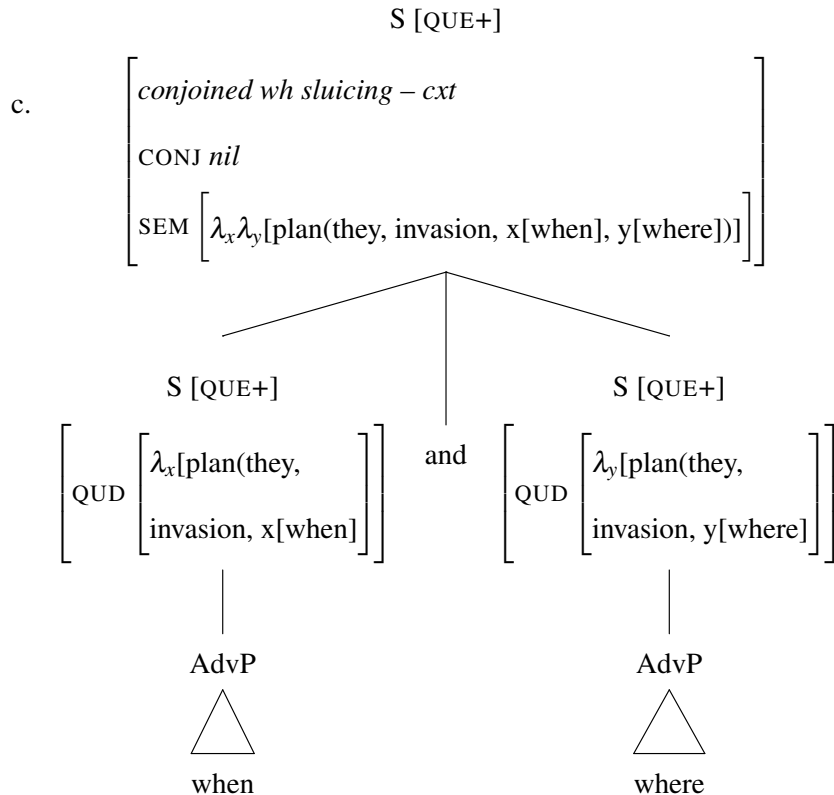


The proposed structure in (5) looks much similar to that of reverse sluicing analysis ([Gianakidou and Merchant, 1998](#)), displaying the two *wh*-remnants are daughter nodes on the same

level that both generate the collective meaning of the mother node formed by the functor ‘*and*’, which conjoins the two conjuncts. In order, the first conjunct is projected into having a propositional meaning by the *Head-Fragment* construction, and also acquires a QUE+ feature. The second remnant undergoes the same process, and the two remnants, now both a sentence, is able to be conjoined by a coordinating conjunction ‘*and*’. Therefore, the propositional meaning of the whole structure is combined into one. The following section will try to account for the *completely*, *partially*, and *not sharing* cases of conjoined *wh*-sluicing, classified in Chapter 4.2.3.

For *completely sharing* instances, the two variables will contribute to the meaning of the whole sentence.

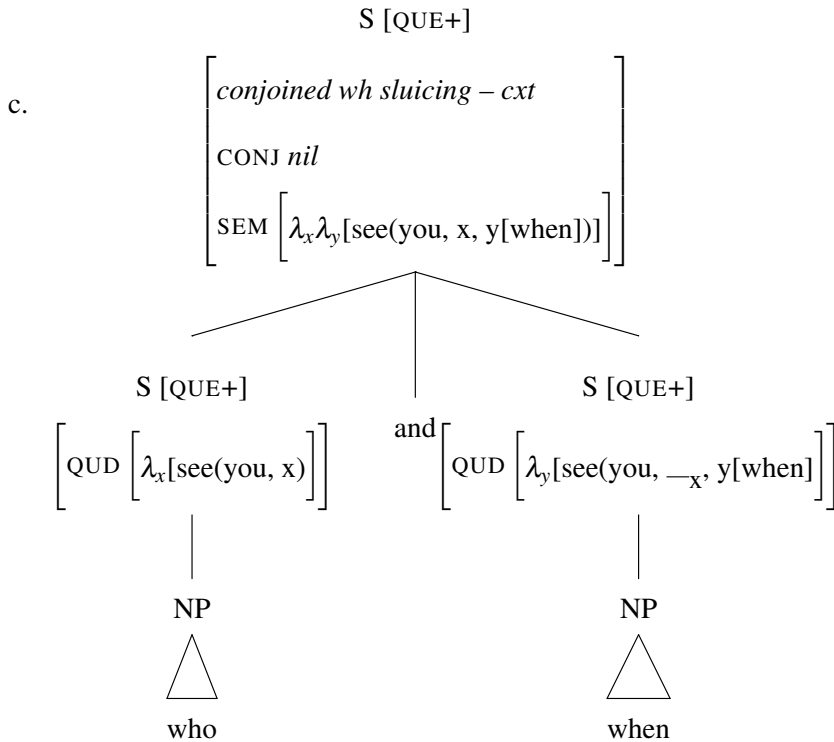
- (6) a. “Word is, they’re planning an invasion of England. And I have a way of finding out **when and where**.” (TV 2002 US/CA)
- b. ...finding out **when** <they’re planning an invasion of England> and **where** <they’re planning an invasion of England>.



The example in (6a) has the putative source clause of the two *wh*-remnants derived from the antecedent, as can be seen in (6b). Each variable is linked to respective adjunct meanings, where the first remnant ‘*when*’ is inquiring the time of the proposition whereas the second remnant ‘*where*’ is asking the place. The cumulative meaning from the *completely sharing* cases like (6) consist of the two variables which can inquire the evoked interrogation simultaneously.

This structure accounting for the *partially sharing* cases shows that the variables can actively constitute the semantic meaning of the sentence, as part of the argument structure.

- (7) a. “You can see characters often at the Magic Kingdom although it’s hard to pin down exactly **who and when**.” (NOW 2020 US)
- b. ...hard to pin down exactly **who_i** <you can see at the Magic Kingdom> and **when** <you can see ____i at the Magic Kingdom>.



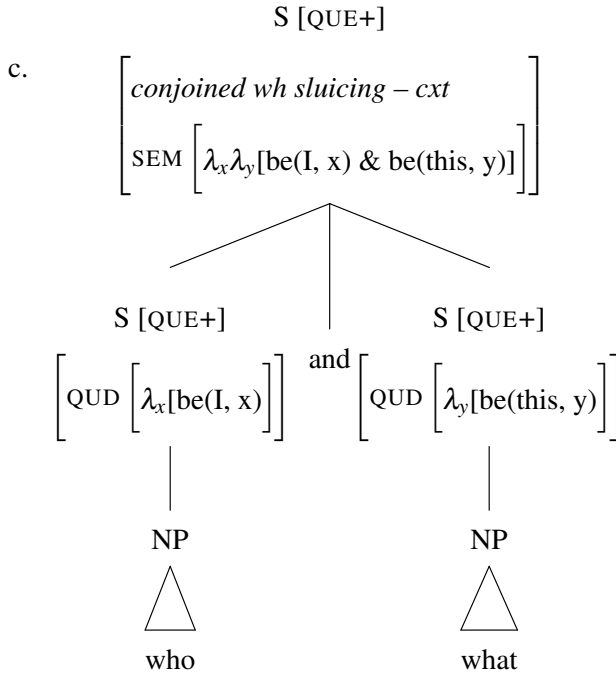
In this case of conjoined *wh*-sluicing, ‘*who*’ accounts for the direct object of the sentence, for its correlate is ‘*characters*’ in the antecedent clause. The adjunct-denoting *wh*-remnant ‘*when*’ is attached at the end, with the index of the *wh*-remnant ‘*what*’ in its putative source clause. One

thing to note is that while the meaning of the first remnant ‘*who*’ as an argument has a complete sentential meaning, the second remnant ‘*when*’ implies that there should be a direct object that would be resolved when met with the QUD of the first remnant.

Lastly, the combined meaning of the *wh*-remnants which are defined respectively can be represented in the *not sharing* case. These were the instances that could not be fully explained through the analyses based on the Minimalist framework. With the *conjoined wh-sludging* construction, these accounts can be analyzed as well. In the case below, the MAX-QUD, the semantic aspect of the sentence is crucial for the projection of each *wh*-remnant.

- (8) a. “His lower lid almost swelled the whole left. Eye closed. I taped his right eye shut. I said, “This will help, " but didn’t say **who** and **what**.” (COCA 2001 FIC)

- b. ...didn’t say **who** <I am> and **what** <this is>.

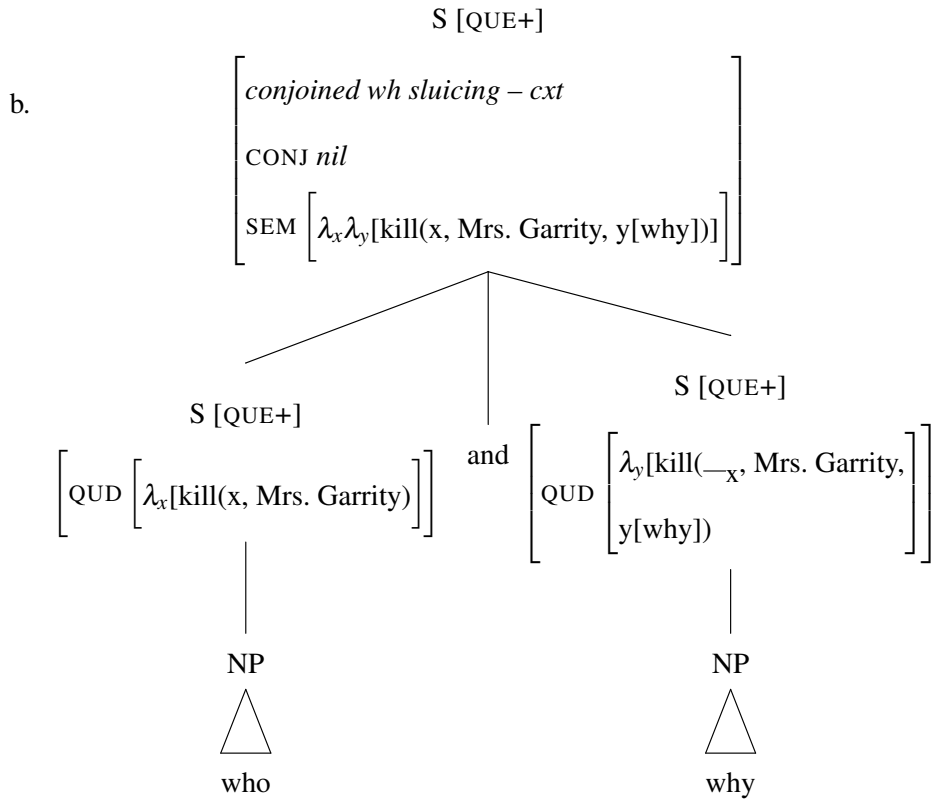


The semantic meaning of the sluiced site derives from the contextual information given previously. In the preceding context, the speaker knows what he is doing to another entity ‘*he*’, but it seems like the speaker did not reveal or clarify who he is to the other person in the context, because

he taped his right eye shut. Therefore, the CONJ value in this case of conjoined *wh*-sluicing will function to coordinate two propositions into one, as can be seen in the SEM value in the structure (8c) above.

The *conjoined wh-sluiding* construction can also explain the preposition and left-branched possessive *wh*-remnant instances, because the construction focuses on the projected sentential meaning of the sluiced site rather than the strict syntactic structure.

- (9) a. “I think I know how Mrs. Garrity was killed. I think I know **who and why**.”
(TV 2016 UK/IE)



In the preposition drop case with voice mismatch as in (9) can show that the intended meaning of the sentence suffice through the voice mismatch phenomenon. Fundamentally, the differences between passive and active structures are related to information packaging, but they still intend to display the same truth value conditions. This could be represented in the structure of *conjoined wh-sluiding* construction, as in (9c). The structure looks similar to that of *partially sharing* case

where the first remnant ‘*who*’ has a complete sentential meaning with all of its required arguments, while that of ‘*why*’ has a implication that it requires an argument from the preceding remnant and that would be fulfilled in once the two propositional meanings are combined into one.

From the analysis given in this thesis, I would like to assert that *conjoined wh-slulcing* is a construction that displays more flexibility in the forming the putative source clause for the remnants. Since the construction is not confined to strict syntactic structural restrictions, it is able to account for unique data that has been mentioned in the previous section. In addition, this structure is able to produce the Single-Pair reading, which is a distinctive semantic characteristic of the construction, for each of the remnants lie in a separate sentence, conjoined by the coordinating conjunction ‘*and*’. In all, I believe this internal structure of *conjoined wh-slulcing* better represents the collected authentic data in the previous section.

Chapter 6

Conclusion

Conjoined *wh*-sluicing in English is an elliptical phenomenon with distinct properties of its own, in comparison with other types of multiple *wh*-constructions such as multiple sluicing and coordinated multiple *wh*-question. Conjoined *wh*-sluicing requires argument structure of the putative source clauses and the antecedent to be identical while also allowing sluicing types of merger and sprouting to be coordinated in four types of combination. Unlike other multiple *wh*-constructions which constrain their argument-denoting *wh*-expressions to certain grammatical function, conjoined *wh*-sluicing does not, which displays the flexibility of the construction. The construction has an insensitivity to locality restrictions, like the ability to express various types of remnants through SWIPING, violate the clausemate condition and repair island violations displays its own features as an elliptical construction. The construction holds semantic properties that distinguishes it from multiple sluicing, in that it restricts the scope of its correlates to the existential reading, generating a Single-Pair interpretation.

This semantic aspect of conjoined *wh*-sluicing is portrayed in the previous analyses of the construction given in the Minimalist framework. There are four analyses, a mono-clause analysis where the two remnants originate from the same clause, was ruled out because it produces a Pair-List interpretation as well. Second, the reverse sluicing analysis views the putative source clause of the second remnant to be copied to that of the first, but since the analysis is limited to ‘*adjunct-adjunct*’ pair, it was not considered to be the internal structure of conjoined *wh*-sluicing.

Next, the bi-clausal bulk sharing analysis is similar to the two preceding analysis, but differ in that the two remnants remain in distinct clauses, while sharing every element in the putative source clause of the second remnant to the first, which poses dominance. Last is the bi-clausal non-bulk sharing analysis, which assumes a multidominance relationship of the two remnants and therefore the putative source clause of the first remnant could be selectively chosen to consist that of the second. However, these analyses as well are not sufficient to account for instances of conjoined *wh*-sluicing where the putative source clause of the two remnants are not identical.

Therefore, a corpora investigation, with six types of corpora from the BNC corpus family was implemented to observe the characteristics of conjoined *wh*-sluicing in an authentic, native setting. There were several findings, which included the observation of four types of remnant pairs, displayed that the construction had a '*adjunct-argument* pair' which had not been mentioned by the previous works. Second, the construction preferred an overt correlate in case of argument-denoting remnant. Due to the '*adjunct-adjunct* pair', which consists 68% of the data with a majority of them sprouting, accounts for the large proportion of preference in the '*covert-covert* pair'. Third, the construction prefers its remnants and antecedent to completely share the putative source clause, but the cases that do not share takes up approximately 35% of the data, which requires to be accounted for. Lastly, the grammatical functions of argument-denoting *wh*-remnants serves as an empirical evidence that the remnant can hold various grammatical functions, such as subject, object, prepositional object, predicative complement, and possessive.

In the discussion section, I discussed some idiosyncratic instances from the dataset, such as the cases which violate the structural parallelism between the putative source clause of the remnants and their antecedent. These cases would require a novel approach that would not assume that putative source clauses of remnants and their antecedent to be identical. The authentic data also revealed that conjoined *wh*-sluicing displays cases of SWIPING while also indicating that there are some issues with accounting for the syntactic identity issues which involve locality constraints. In the end, a covarying collexeme analysis was conducted to observe whether some remnant pairs had strong attraction between each other. While most repelled, there were pairs that attracted, many of which involves one argument *wh*-remnant and an adjunct one.

Finally, this thesis strives to discuss a CxG, HPSG based approach to conjoined *wh*-sluicing in English, which does not suppose the internal structure between the putative source clause and the antecedent to be completely sharing. On the other hand, *conjoined wh-sluicing* construction proposed in this thesis with the Head-Fragment Construction (Kim and Nykiel, 2020), will be able to resolve syntactic identity issues which are raised in the corpora dataset, by assuming that a single *wh*-remnant can consist of a daughter node which can be directly projected to a acquire a sentential meaning, to be conjoined by a functor ‘*and*’. This structure better represents the data collected in section 4, and better accounts for the syntactic and semantic properties of the construction.

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국문초록

본 연구의 목적은 영어에서 다중 병렬 *wh*-sluicing 현상을 조사하는 것이다. 이는 다중 sluicing 및 병렬 다중 *wh*-질문과 같은 다른 다중 *wh*-구조와 비교하여 고유한 특성을 가진 생략 구문이다. 다중 병렬 *wh*-sluicing은 국부 제약(locality constraint)에 둔감하여 SWIPING 현상을 통해 다양한 잔여구(remnant)들을 생성하고, 절 동료 조건(clausemate condition)을 위반하며, 섬 제약 위반(island constraint violation)을 허용함으로써 독특한 생략 언어 현상으로 특징지어 진다. 의미적으로 다중 병렬 *wh*-sluicing은 상관어의 양화사를 existential의 범위로 제한하여 Single-Pair의 해석만 가능하게 된다.

기존의 최소주의 틀 내에서의 분석, 즉 단일 절 (mono-clause), 역 sluicing (reverse sluicing), 이중 절 (bi-clause)을 통한 분석은 잔여구와 선행절 (antecedent clause)이 동일하지 않은 경우에 대한 접근이 불가하다는 것을 여러 예시를 통해 제안하였다. 이를 분석하기 위해 BNC 코퍼스 계열의 6개 코퍼스를 활용한 코퍼스 조사가 실시되었고, 해당 구문에 대한 특징적인 요소들을 통해 구분하였다.

논의 섹션에서는 데이터에서 몇 가지 특이 사례를 다루며 해당 구문에 대한 새로운 접근법의 필요성을 제시했다. 실제 데이터는 섹션 2에서 제시된 몇 가지 기본 특성을 강조했으며, SWIPING 사례와 국부제약을 포함한 통사적인 면에서 구문의 해석이 어려운 부분을 제시했다. Covarying collexeme 분석으로 데이터의 질적 분석을 진행하며 하나의 adjunct 잔여구와 하나의 argument 잔여구를 포함하는 다양한 잔여구 쌍 사이에서의 attraction과 repulsion을 나타내며, 다중 병렬 *wh*-sluicing 구문 안에서 특정 잔여구들 사이에 preference가 있다는 데이터를 제시하였다.

궁극적으로 이 학위 논문은 다중 병렬 *wh*-sluicing에 대한 CxG 및 HPSG를 통한 접근 방식을 제안한다. 해당 접근 방식은 코퍼스 데이터에서 구문이 통사적으로 어려움들을 해소하고, 각 잔여구가 하나의 절의 의미를 가지며 구문의 의미는 두 개의 잔여구가 병렬로 구성되어 두 절의 의미를 하나의 복합적인 절의 의미로 해석이 가능할 수 있다는 점을 시사한다.

주요 용어: 슬루싱 (sluicing), 다중 슬루싱 (multiple sluicing), 병렬 (coordination), 생략 현상 (ellipsis), Construction-based Grammar